

GRAVITATIONAL DUALS FROM EQUATIONS OF STATE

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NEW INSIGHTS IN BLACK HOLE PHYSICS FROM HOLOGRAPHY

INSTITUTO DE FÍSICA TEÓRICA (IFT), MADRID

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UNIVERSITAT DE
BARCELONA



Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA



EXCELENCIA
MARÍA
DE MAEZTU
2020 - 2024



MINISTERIO
DE CIENCIA, INNOVACIÓN
Y UNIVERSIDADES



AGENCIA
ESTATAL DE
INVESTIGACIÓN

THE PEOPLE

BASED ON:
GRAVITATIONAL DUALS FROM EQUATIONS OF STATE
2403.14763 [HEP-TH], 10.1007/JHEP07(2024)087



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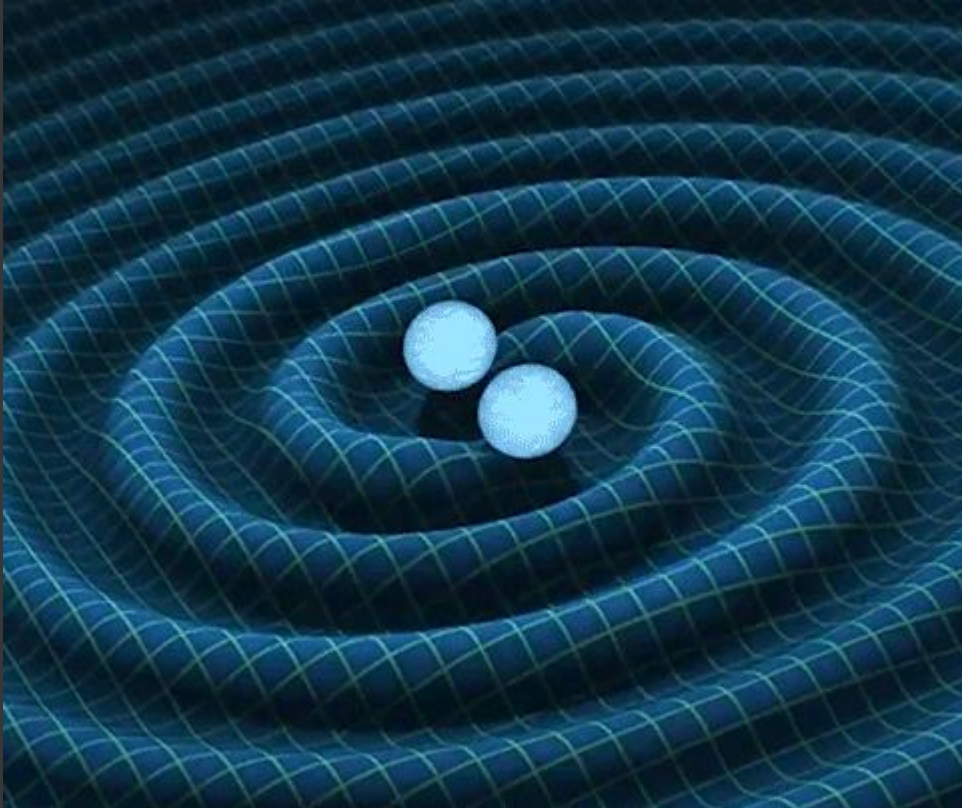


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NEW WINDOWS TO THE UNIVERSE

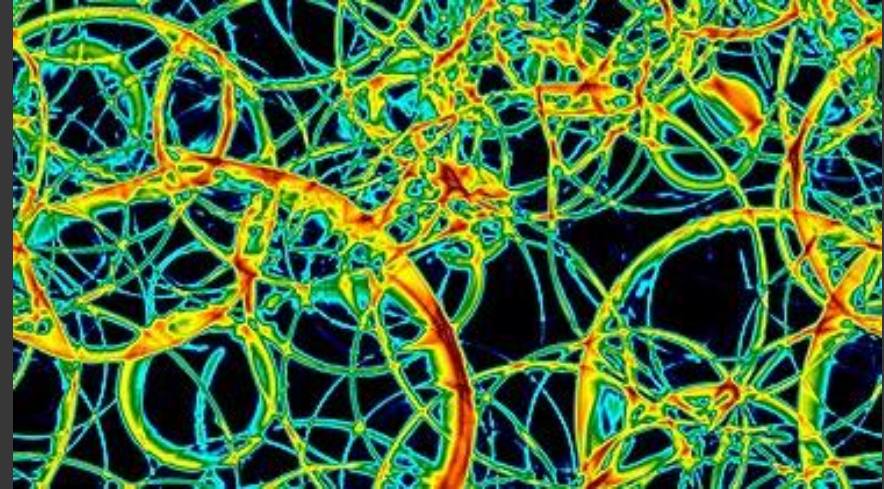


- Strong regime of General Relativity
- Properties of strong-interacting quantum matter

RELEVANT SCENARIOS



Neutron star mergers
SM matter



Phase transition early Universe
BSM matter

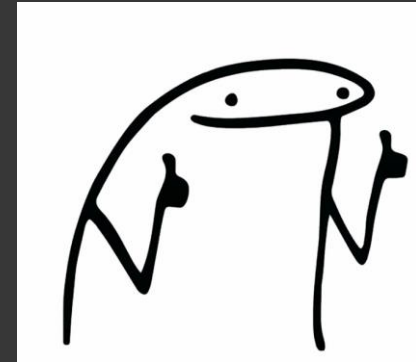
Strongly couple Quantum matter + Out of
Equilibrium



HOLOGRAPHY

WHY HOLOGRAPHY?

- Build the QFT properties from first principles
- Real time dynamics solving Einstein equations in $4+1$ AdS spacetime
- Very difficult to get the $4+1$ AdS spacetime from the QFT properties



Good!

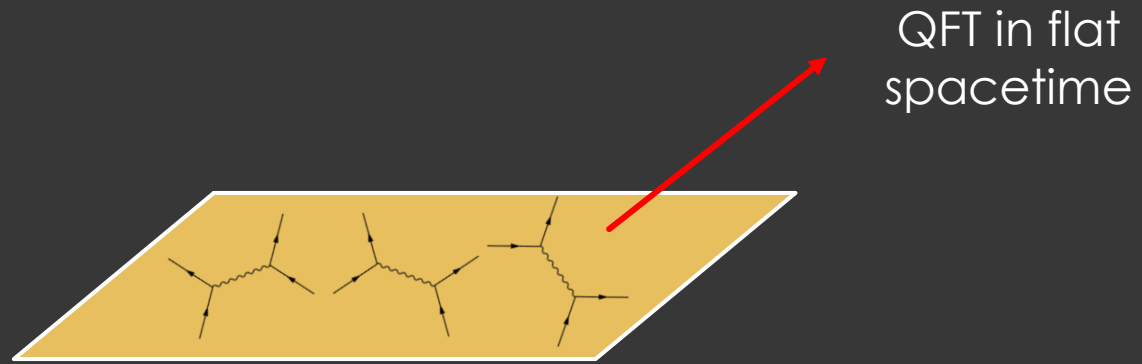


Bad
but...

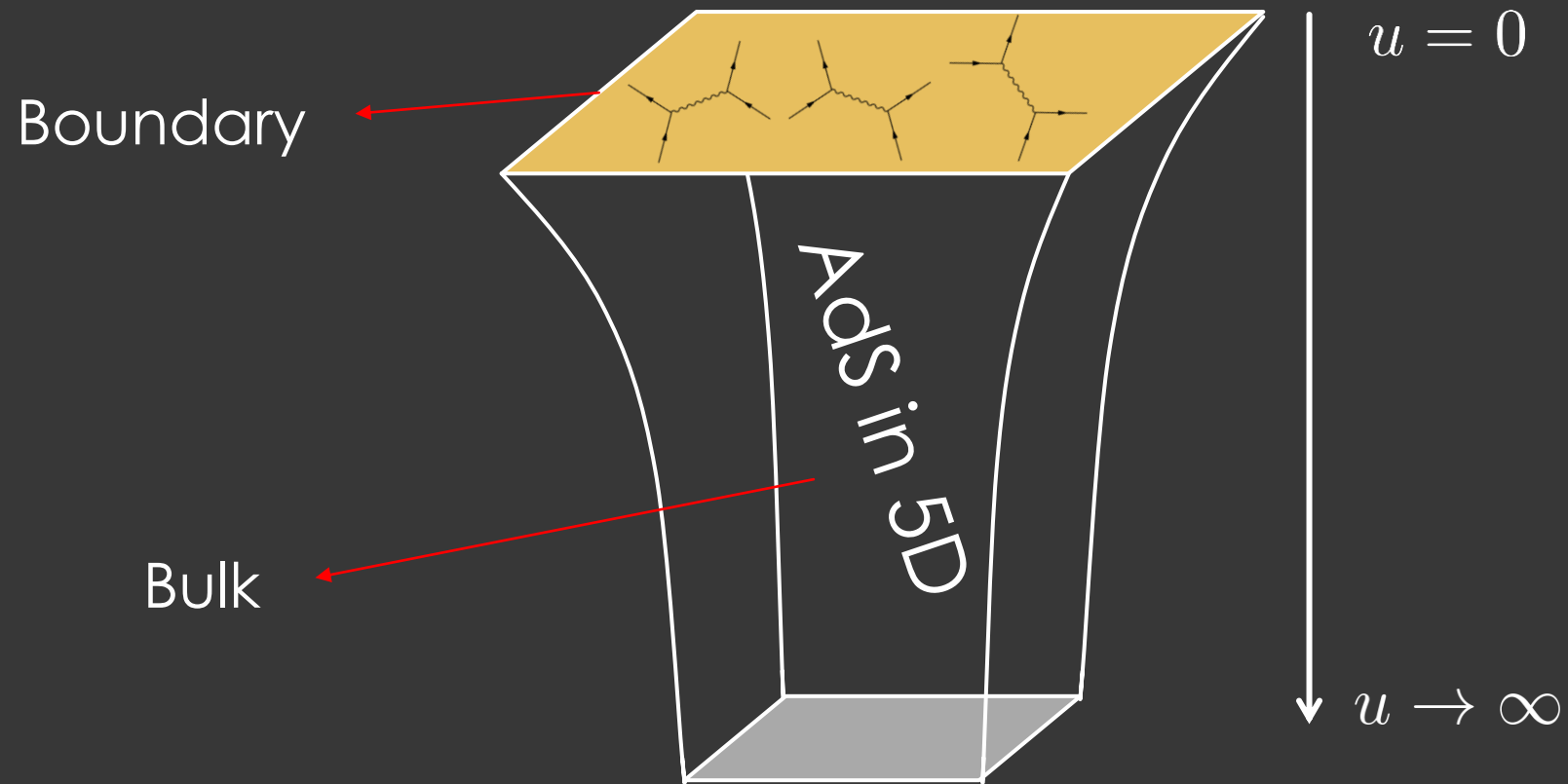
OUTLINE

1. HOLOGRAPHY IN A NUTSHELL: PRESENTING THE MODEL
2. PHYSICS INFORMED NEURAL NETWORKS (PINNs)
3. RESULTS
4. CONCLUSIONS

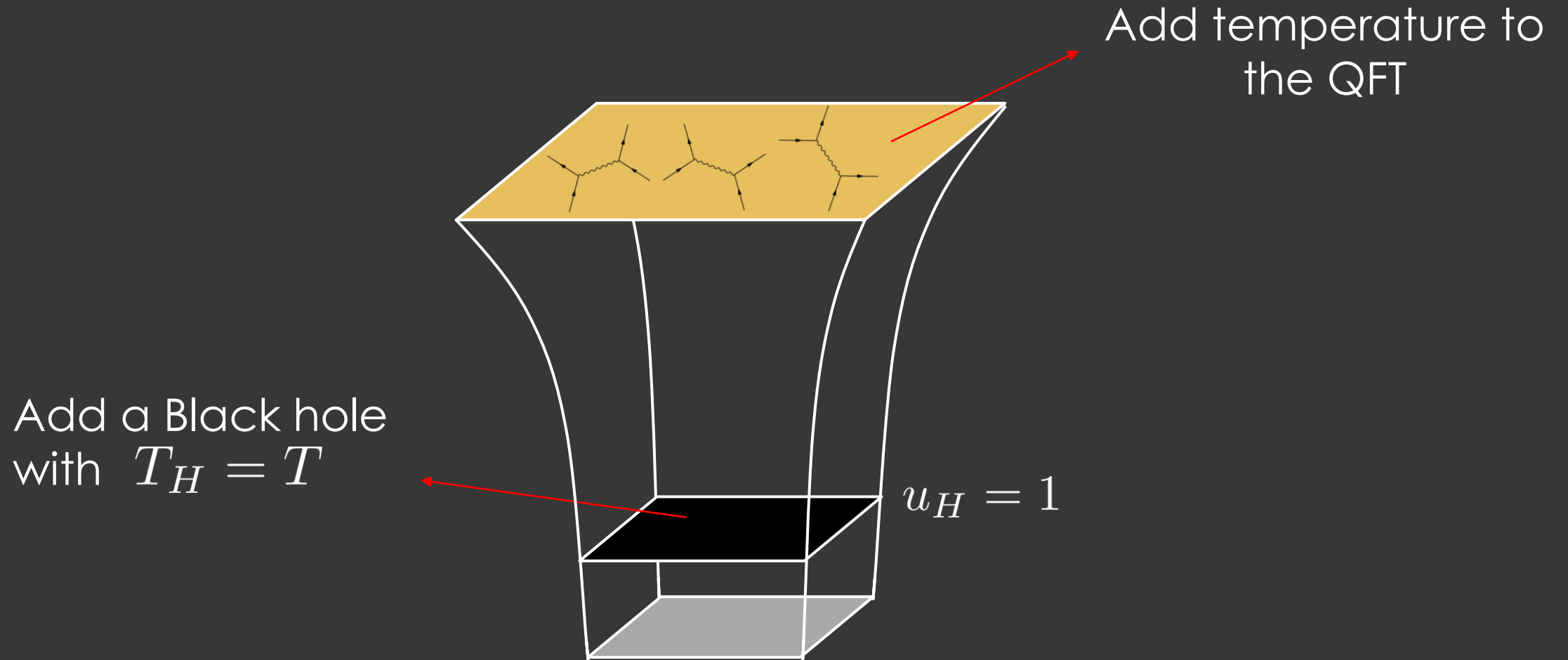
HOLOGRAPHY IN A NUTSHELL



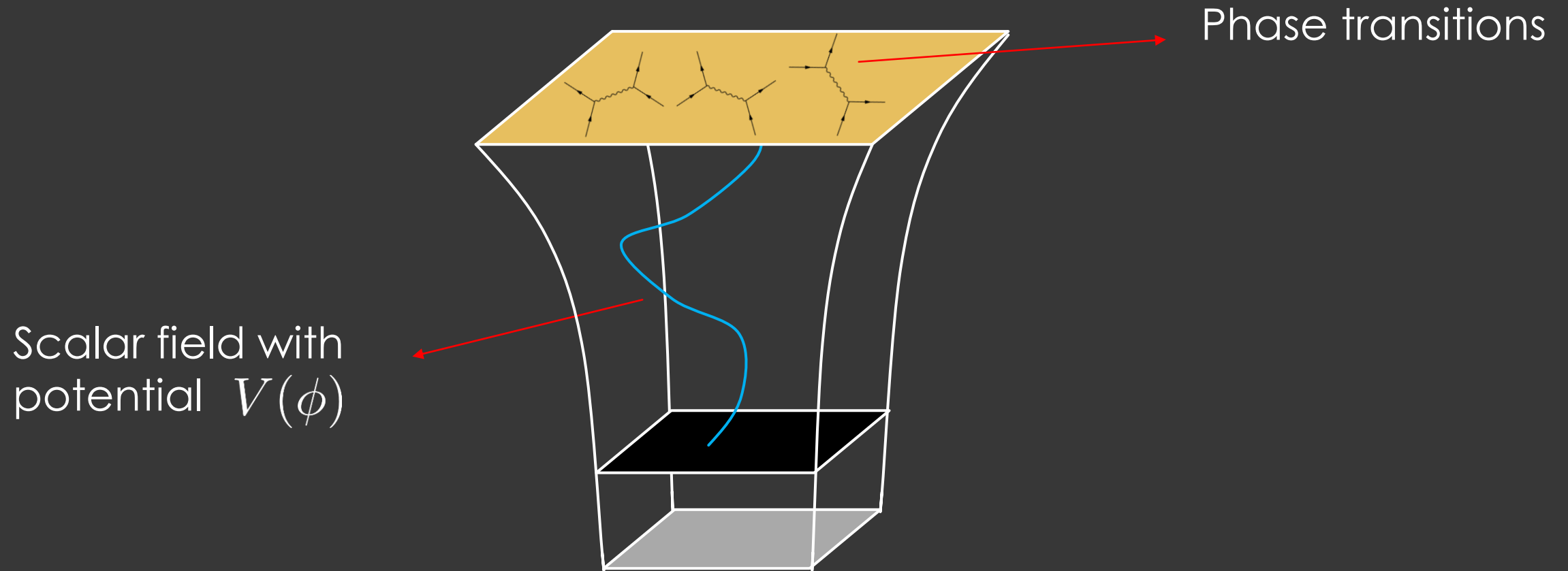
HOLOGRAPHY IN A NUTSHELL



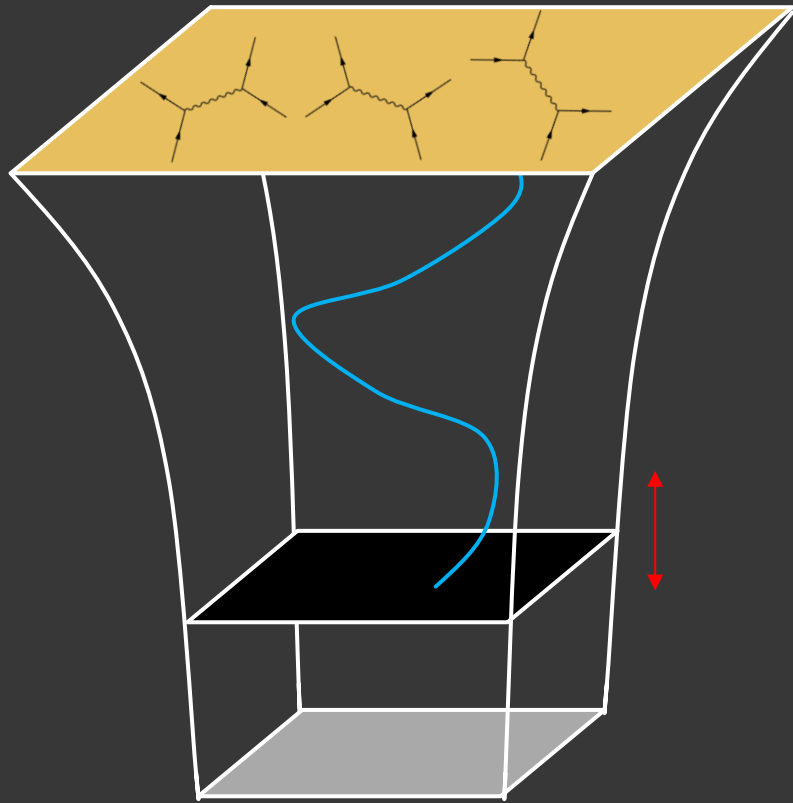
HOLOGRAPHY IN A NUTSHELL



HOLOGRAPHY IN A NUTSHELL



HOLOGRAPHY IN A NUTSHELL

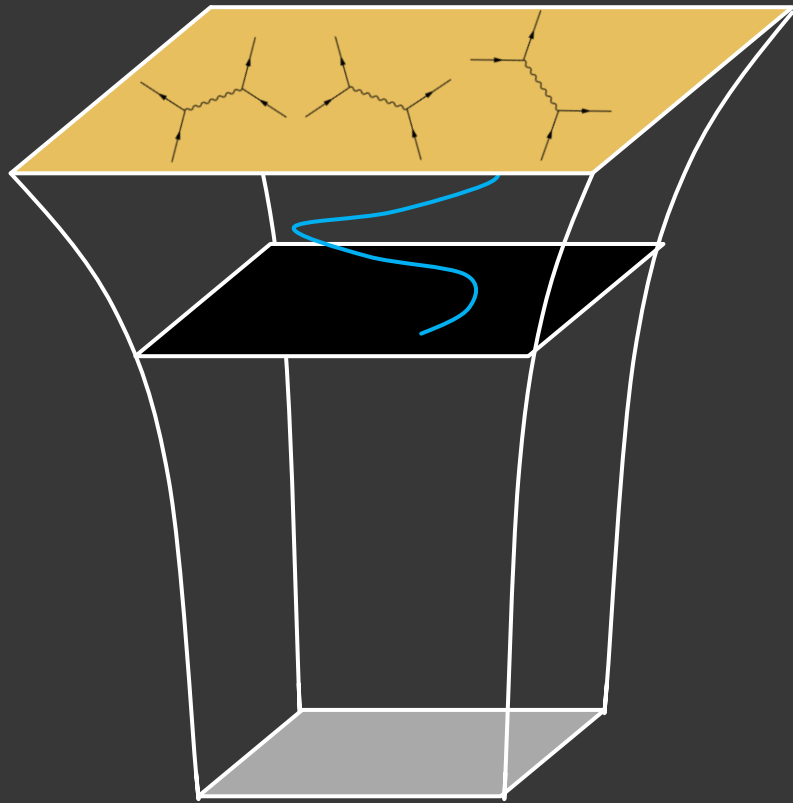


1. Construct different black brane solutions
2. Solve the EE+KG

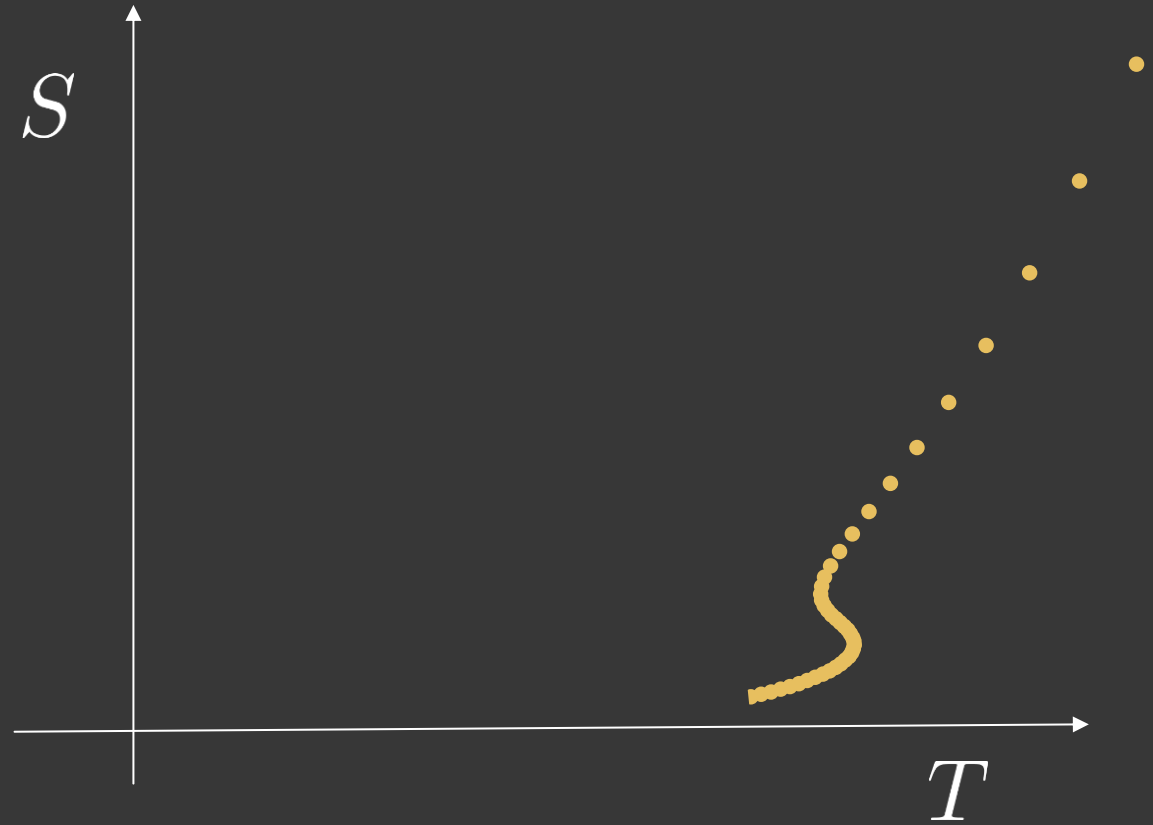
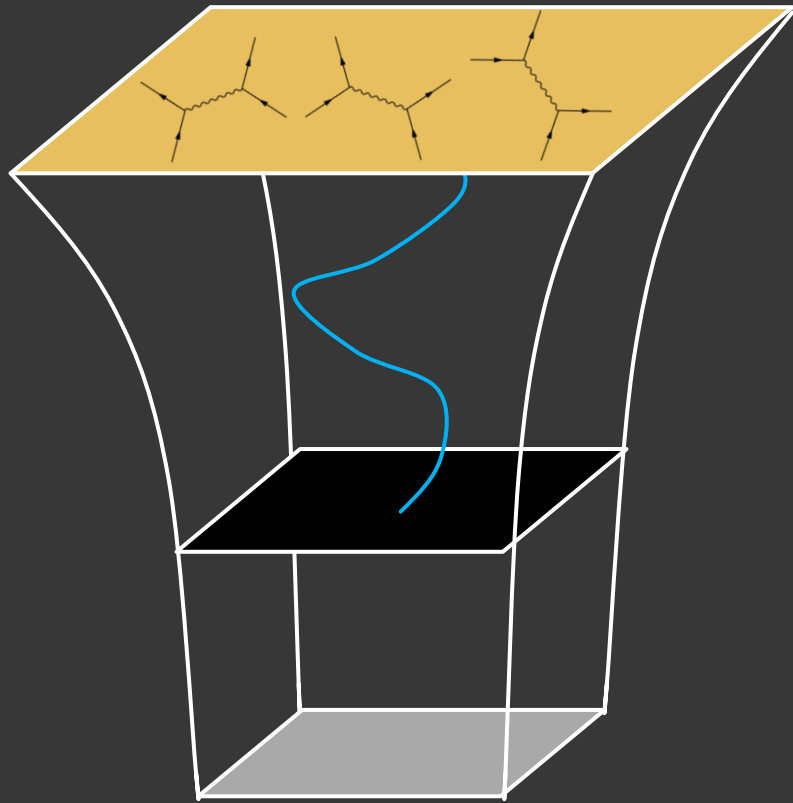
$$S = \frac{2}{\kappa_5^2} \int dx^5 \sqrt{-g} \left[\frac{1}{4} R - \frac{1}{2} (\nabla \phi)^2 - V(\phi) \right] ,$$

3. Read off the thermodynamics as b.c.

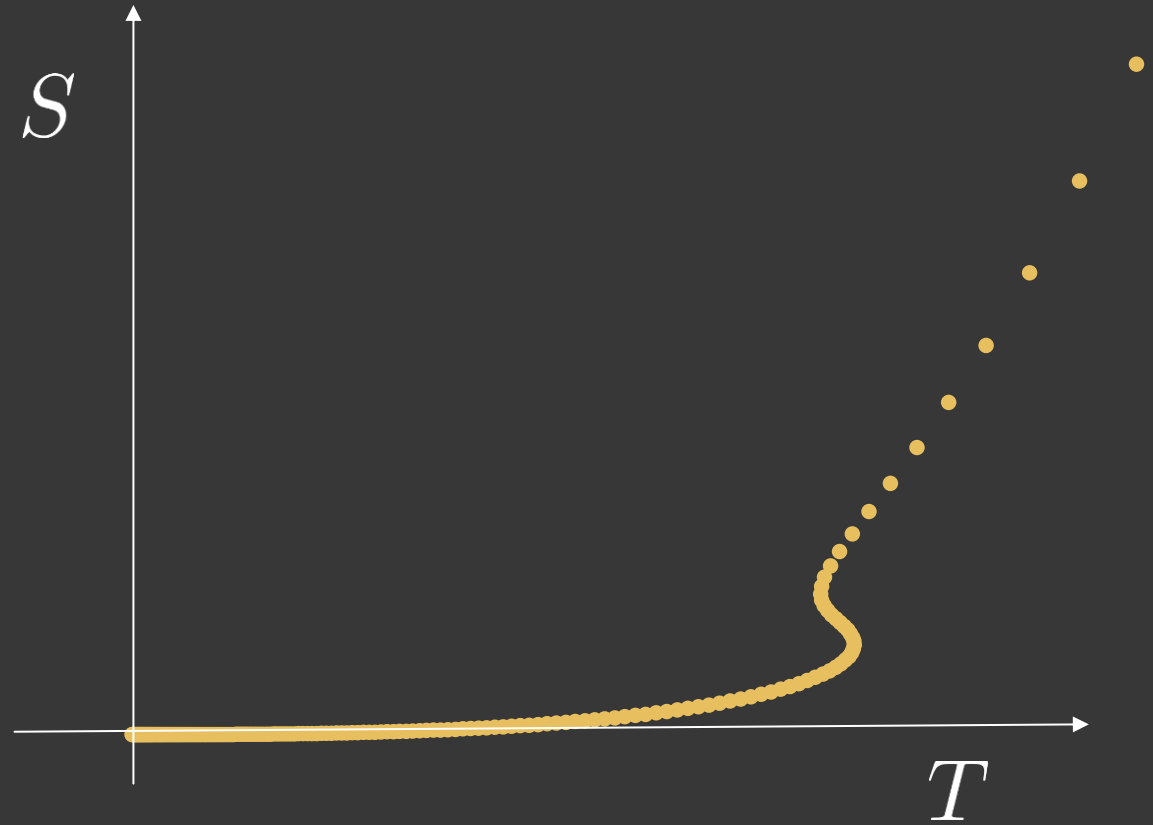
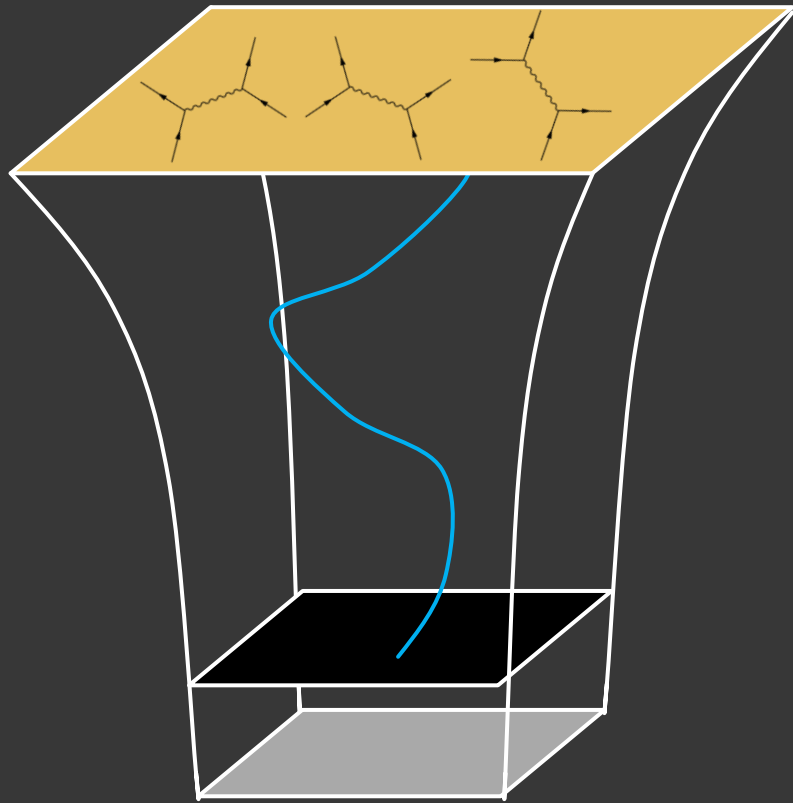
HOLOGRAPHY IN A NUTSHELL



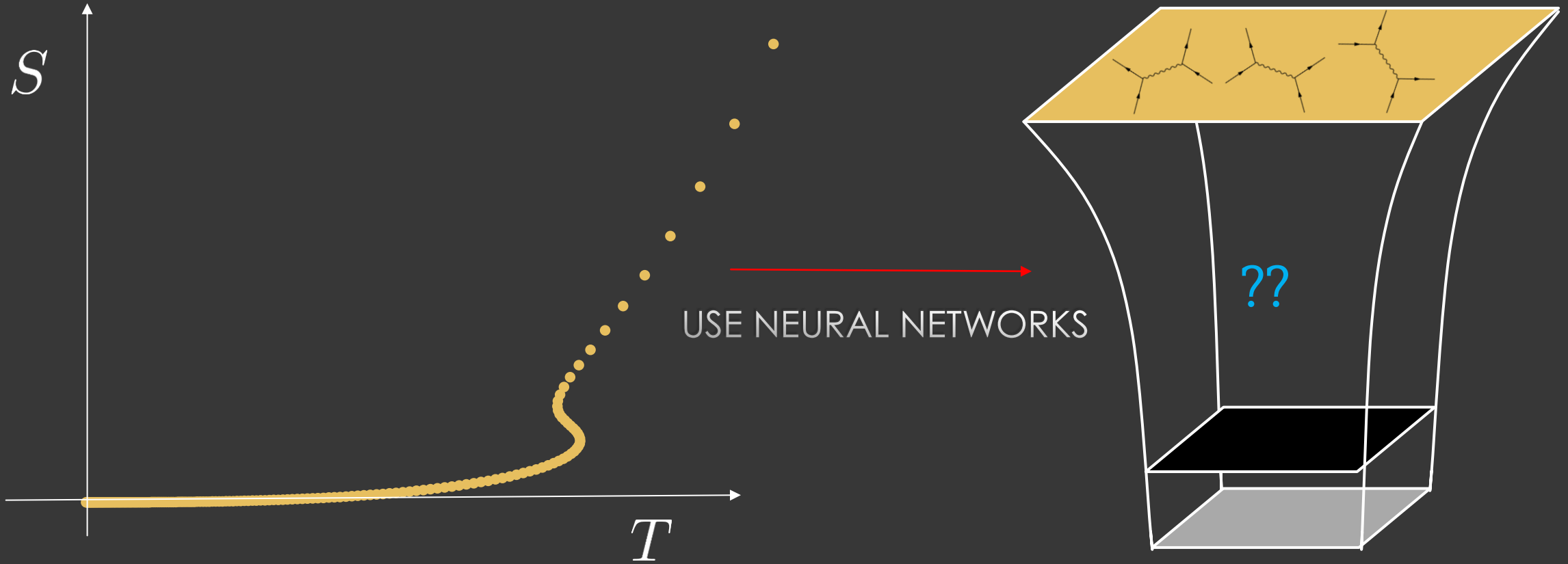
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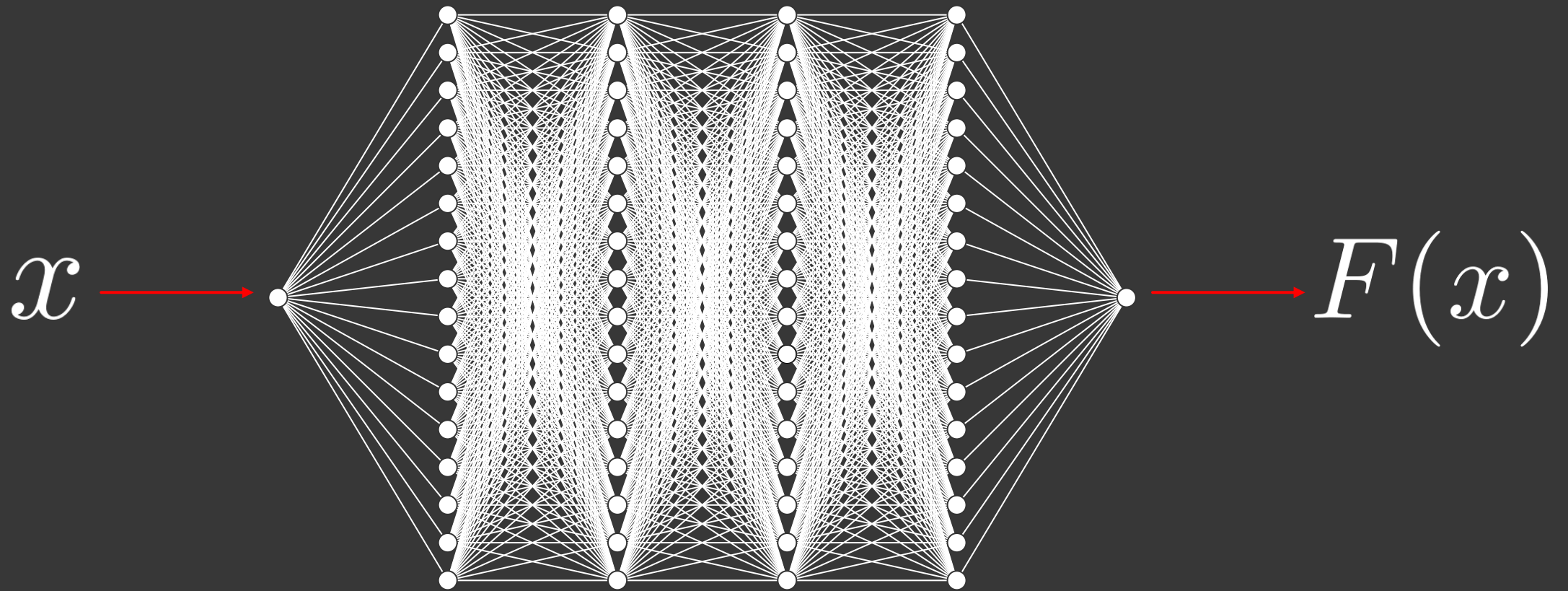
HOLOGRAPHY IN A NUTSHELL



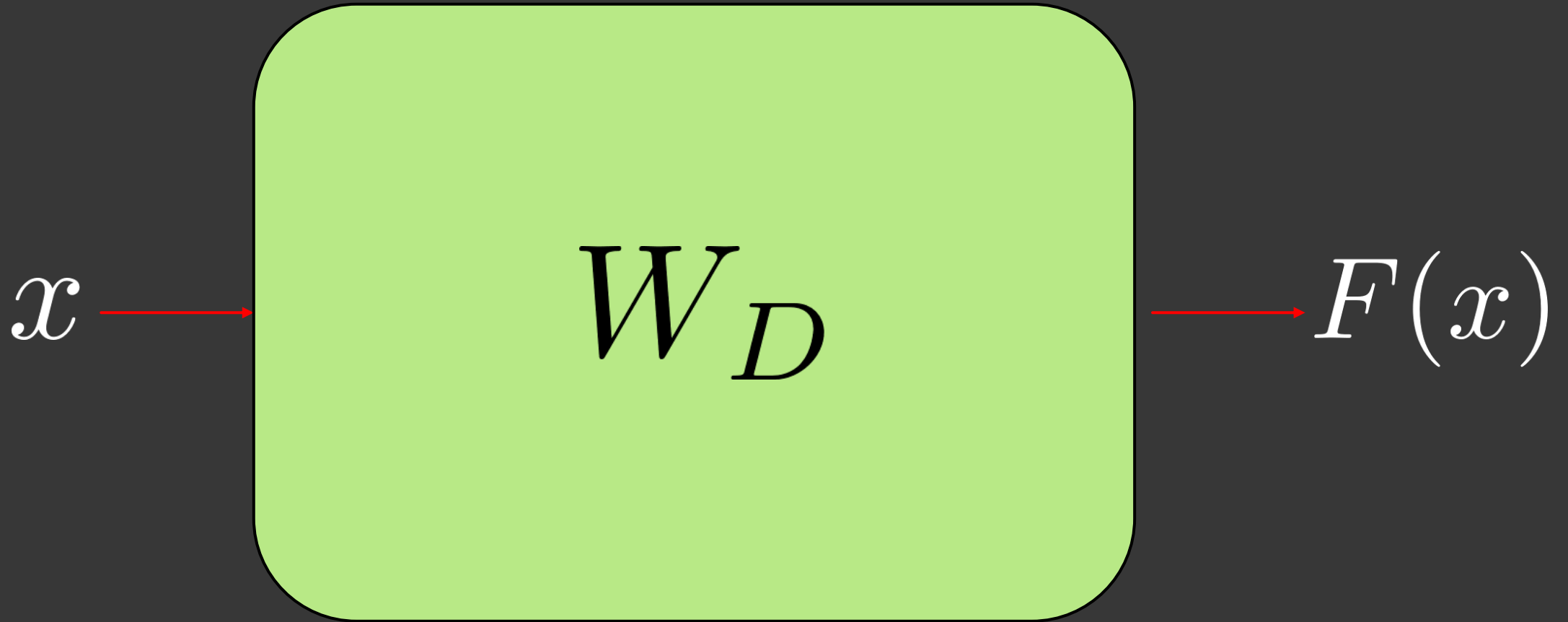
HOLOGRAPHY IN A NUTSHELL



PHYSICS INFORMED NEURAL NETWORKS



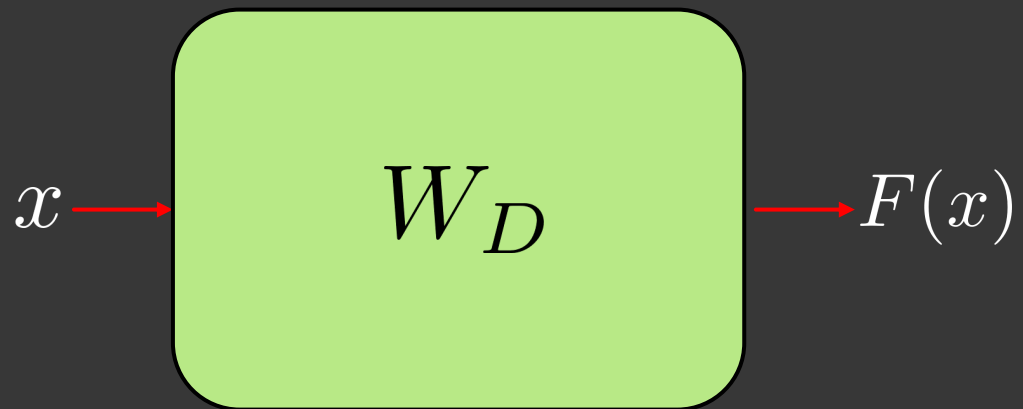
PHYSICS INFORMED NEURAL NETWORKS



PHYSICS INFORMED NEURAL NETWORKS

[Dissanayake, M and
Phan-Thien, N. 1994]

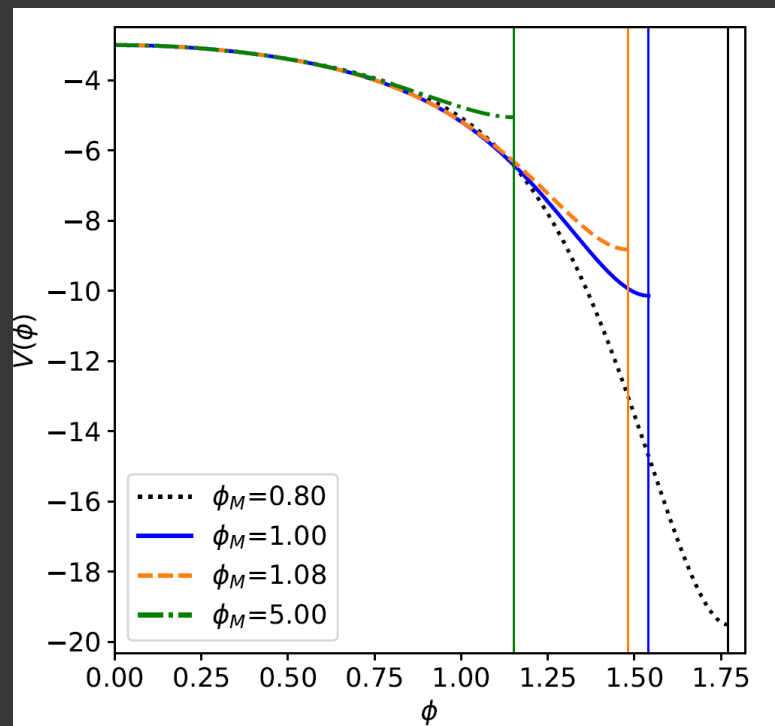
[Lagaris et al. 1997]



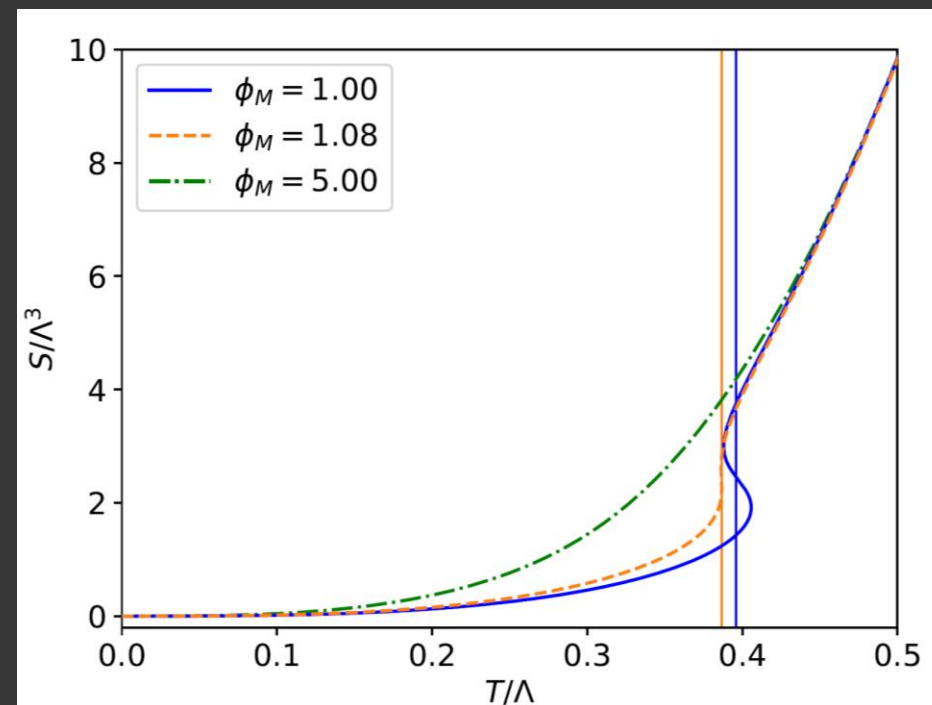
- Use a NN for each unknown function
- BC fulfilled by construction
- Train for different BC at once
- PINNs loss function is defined as the residuals of the differential equations.

[Chen et al. 2020]

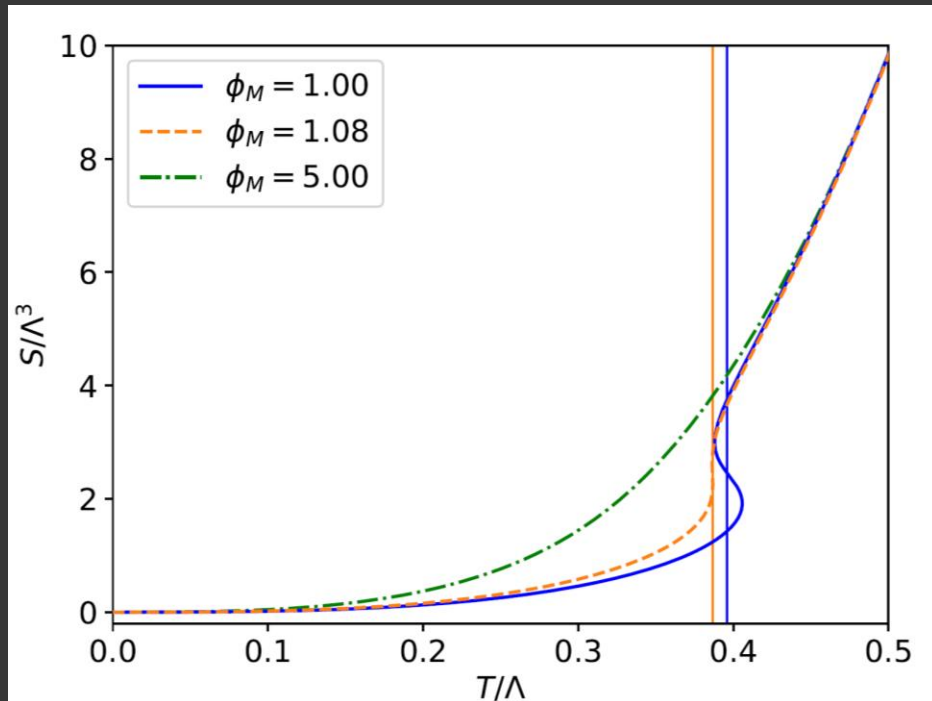
RESULTS



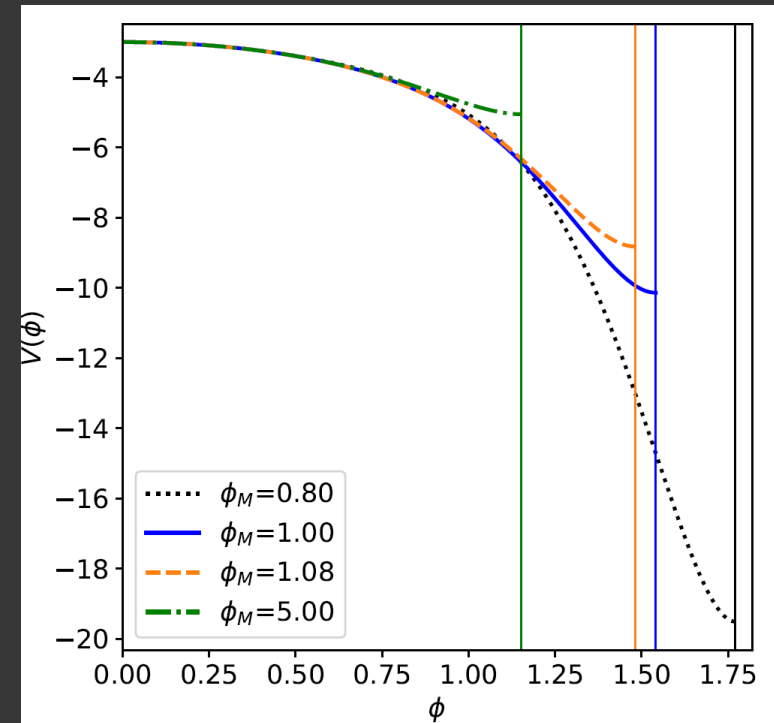
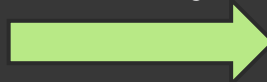
Direct problem



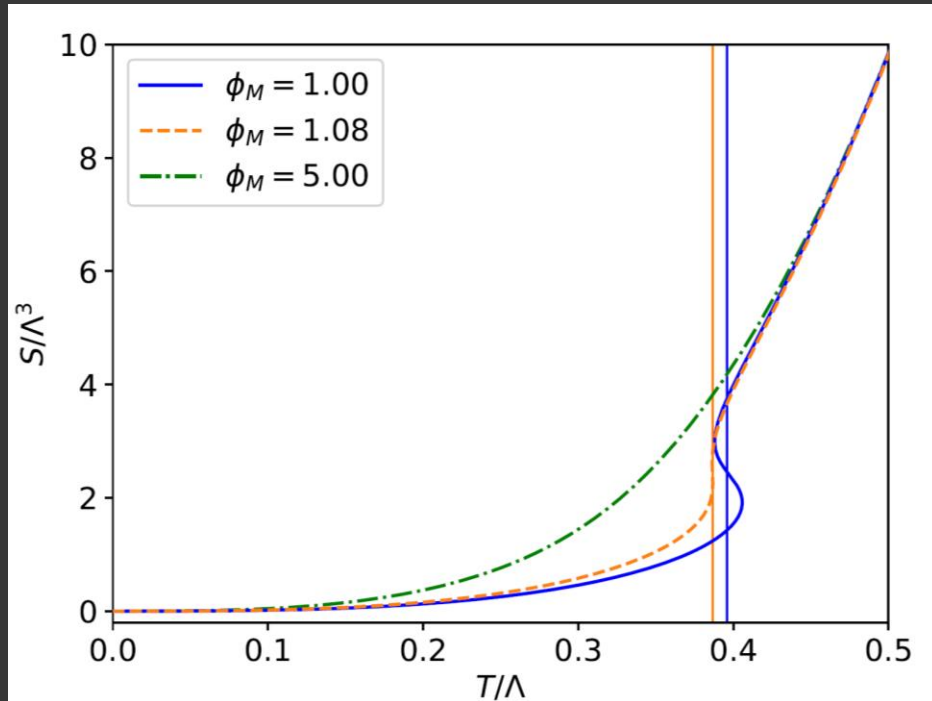
RESULTS



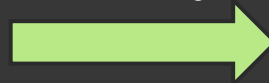
Inverse problem
PINNs



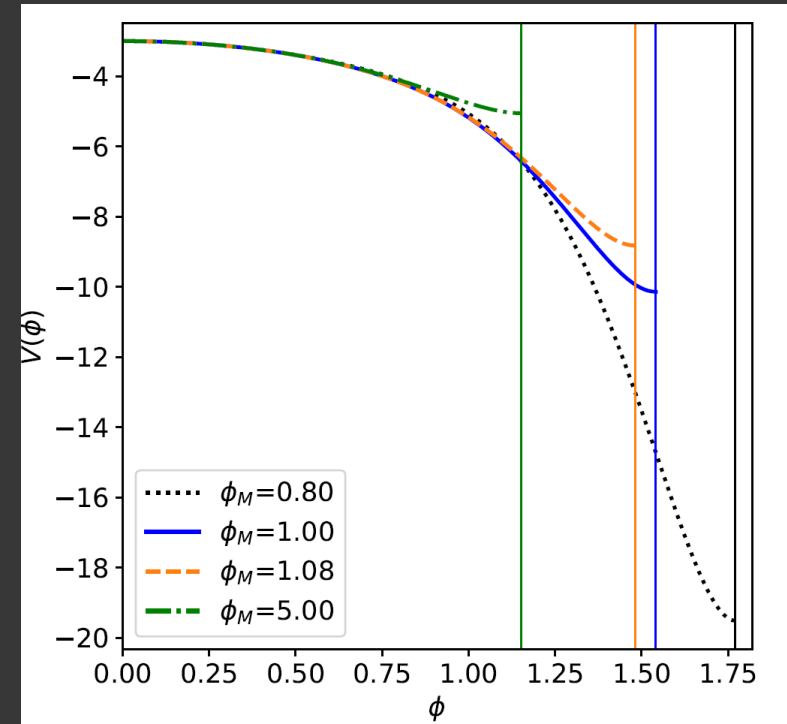
RESULTS



Inverse problem
PINNs

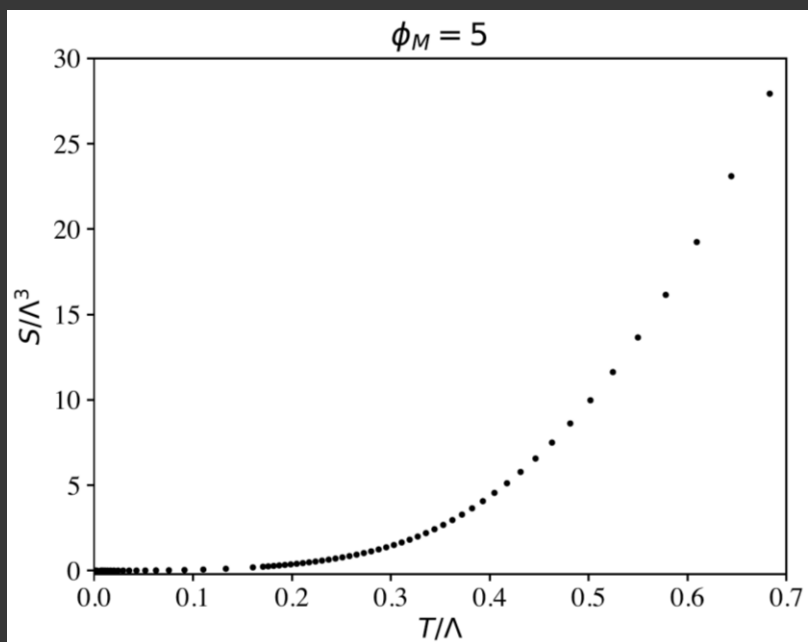


Direct problem
Compare $S(T)$

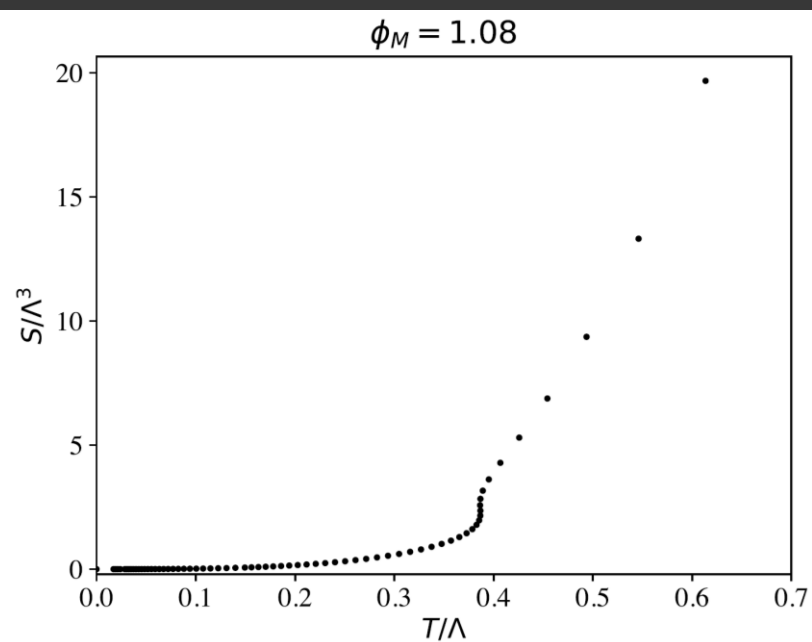


RESULTS

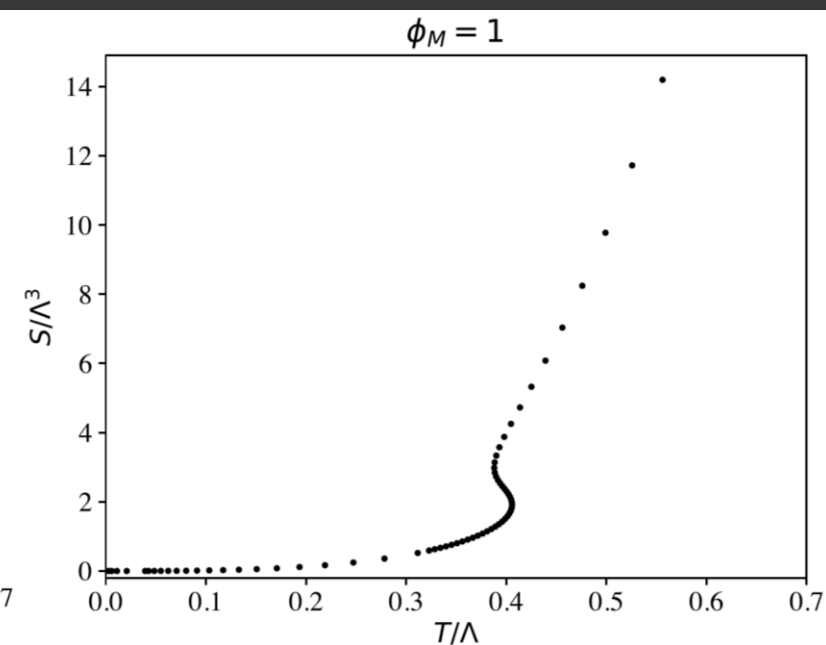
Crossover



Second Order

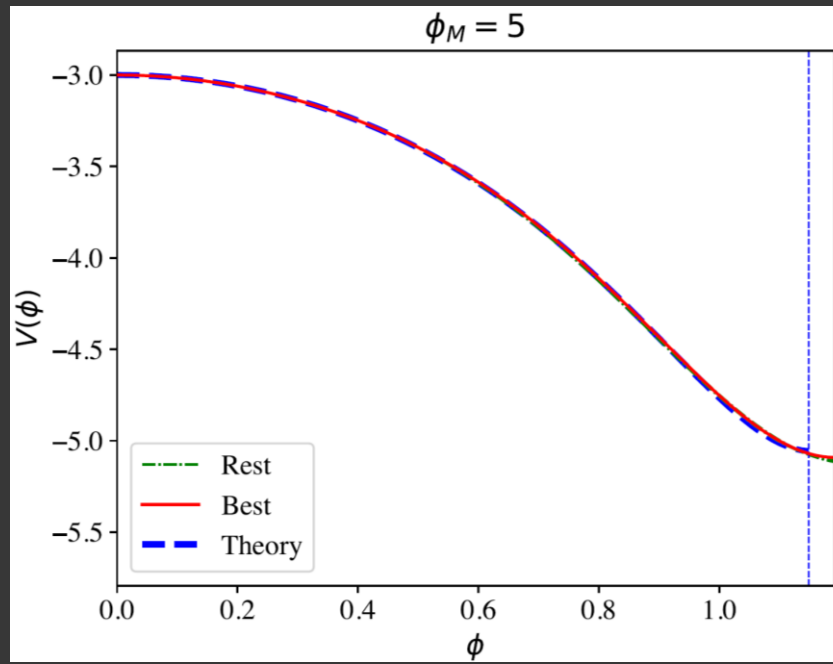


First Order

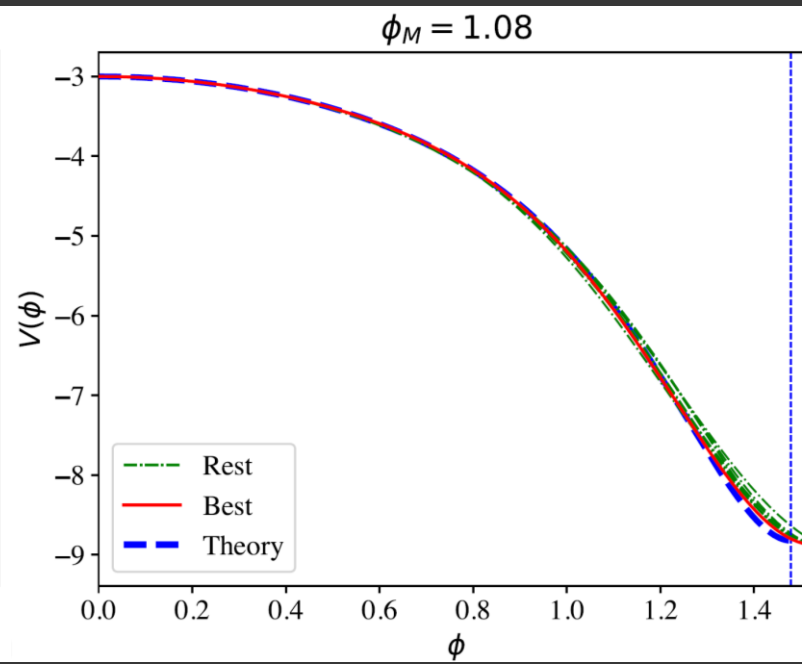


RESULTS

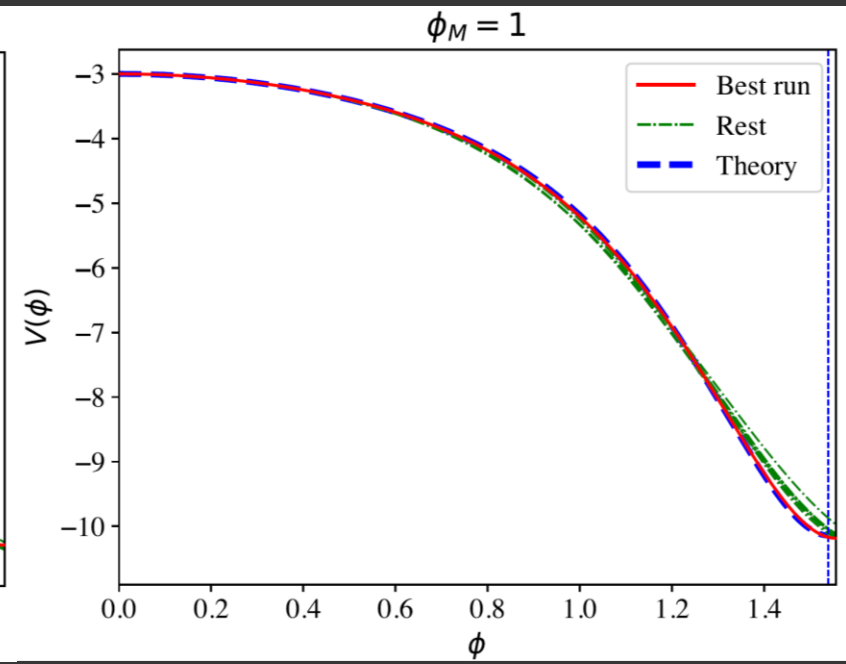
Crossover



Second Order

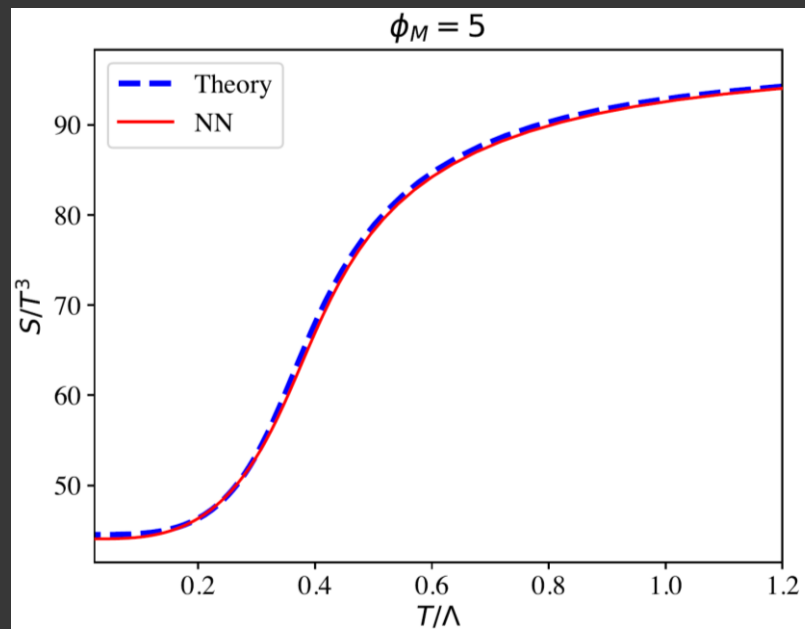


First Order

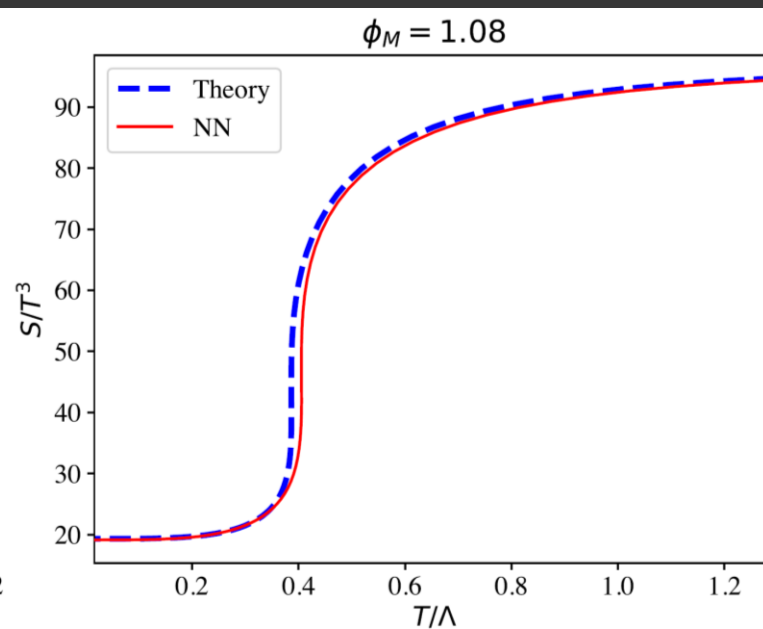


RESULTS

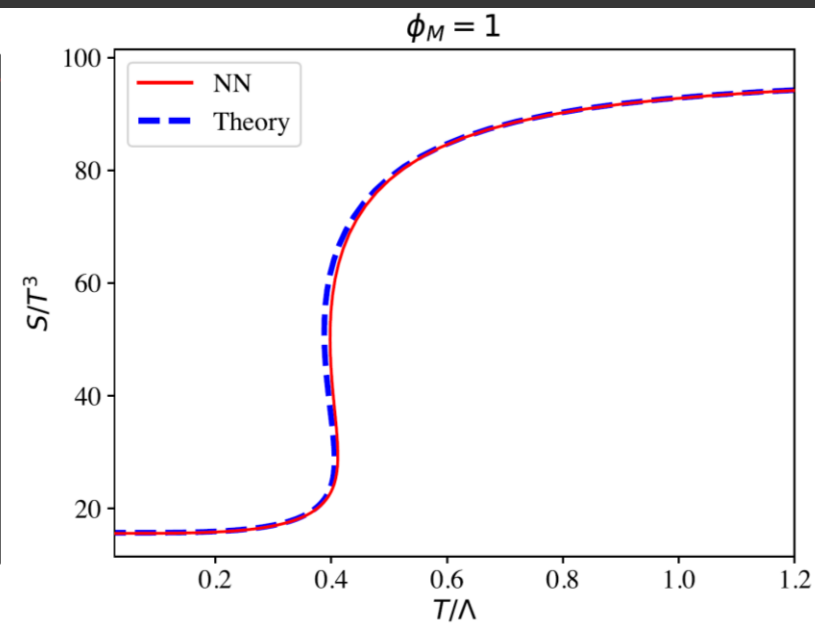
Crossover



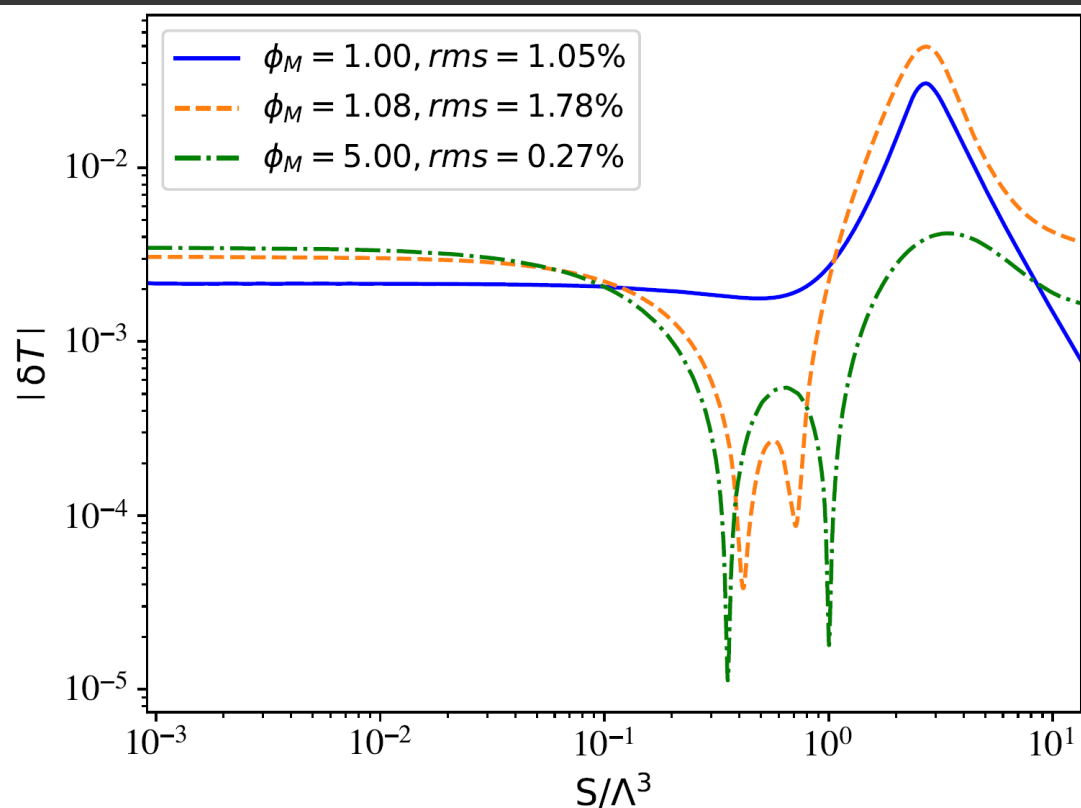
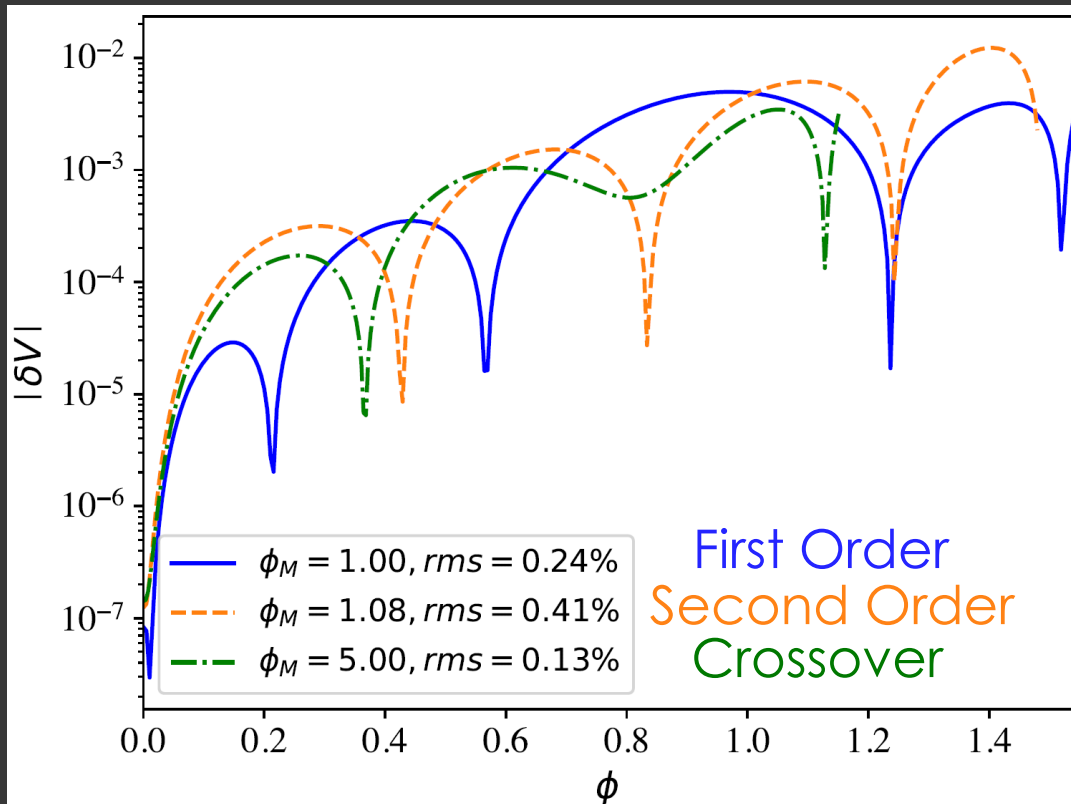
Second Order



First Order



RESULTS



$$|\delta V(\phi)| = \left| \frac{V_{\text{theory}}(\phi) - V_{\text{PINN}}(\phi)}{V_{\text{theory}}(\phi)} \right|$$

$$|\delta T(S)| = \left| \frac{T_{\text{input}}(S) - T_{\text{PINN}}(S)}{T_{\text{input}}(S)} \right|$$

CONCLUSIONS AND FUTURE DIRECTIONS

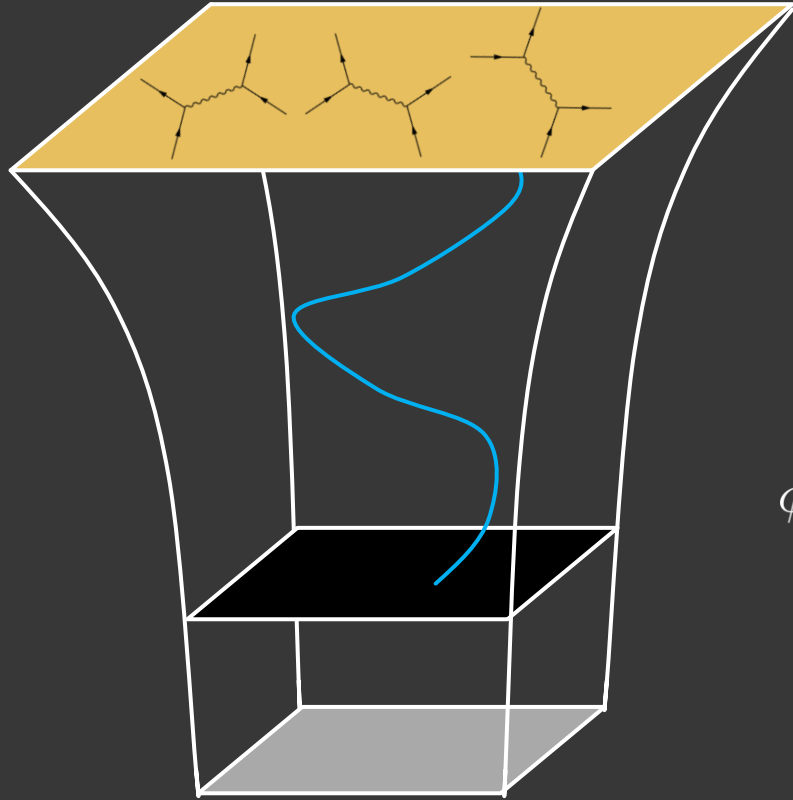
- It is possible to recover the bulk gravitational theory using data from the boundary theory. This is a very non-trivial result in the field of Holography.
- Generalizable to many situations: several scalar fields, gauge fields, QCD equation of state data...
- The setup is not good enough to solve the stiffest cases: false vacuum decay, naked singularity in the bulk... —→ Pau's talk!
- Method generalizable to other inversion problem involving highly non-linear differential equations, i.e. hydrodynamics.
- How are the NNs encoding the RG flow? Could it give us some hints about the scaling of solutions with the energy? —→ Turbulence

THANK YOU!

BIBLIOGRAPHY

- ABBOTT, B., ABBOTT, R., ABBOTT, T., ABERNATHY, M., ACERNESE, F., ACKLEY, K., . . . ZWEIZIG, J. (11 DE FEBRUARY DE 2016). OBSERVATION OF GRAVITATIONAL WAVES FROM A BINARY BLACK HOLE MERGER. *PHYSICAL REVIEW LETTERS*, 116. DOI:10.1103/PHYSREVLETT.116.061102
- CHEN. (2020). NEURODIFFEQ: A PYTHON PACKAGE FOR SOLVING DIFFERENTIAL EQUATIONS WITH NEURAL NETWORKS. *JOURNAL OF OPEN SOURCE SOFTWARE*, 5, 1931. DOI:10.21105/JOSS.01931
- DISSANAYAKE, M., & PHAN-THIEN, N. (MARCH DE 1994). NEURAL-NETWORK-BASED APPROXIMATIONS FOR SOLVING PARTIAL DIFFERENTIAL EQUATIONS. *COMMUNICATIONS IN NUMERICAL METHODS IN ENGINEERING*, 10, 195-201. DOI:10.1002/cnm.1640100303
- LAGARIS, I., LIKAS, A., & FOTIADIS, D. (AUGUST DE 1997). ARTIFICIAL NEURAL NETWORK METHODS IN QUANTUM MECHANICS. *COMPUTER PHYSICS COMMUNICATIONS*, 104, 1-14. DOI:10.1016/s0010-4655(97)00054-4

HOLOGRAPHY EQUATIONS



$$ds^2 = -Adt^2 + \Sigma^2(dx^2 + dy^2 + dz^2) - \frac{2}{u^2} dt du$$

$$\Sigma'' + \frac{2}{u}\Sigma' + \frac{2}{3}\Sigma\phi'^2 = 0 ,$$

$$A'' + \frac{8}{3u^4}V(\phi) + A' \left(\frac{2}{u} + \frac{3\Sigma'}{\Sigma} \right) = 0 ,$$

$$\phi'' - \frac{1}{u^4 A} \frac{\partial V(\phi)}{\partial \phi} + \phi' \left(\frac{2}{u} + \frac{A'}{A} + 3\frac{\Sigma'}{\Sigma} \right) = 0 ,$$

$$A' + 2A\frac{\Sigma'}{\Sigma} - \frac{2\Sigma}{3\Sigma'} \left(A\phi'^2 - \frac{2}{u^4}V(\phi) \right) = 0 ,$$

DIRECT PROBLEM DETAILS

$$\begin{aligned}A(u) &= \frac{1}{u^2} + \cdots , \\ \Sigma(u) &= \frac{1}{u} + \cdots , \\ \phi(u) &= \Lambda u + \cdots ,\end{aligned}$$

$$\begin{aligned}A(u) &= A_1(u_H - u) + \cdots , \\ \Sigma(u) &= \Sigma_0 + \Sigma_1(u_H - u) + \cdots , \\ \phi(u) &= \phi_H + \phi_1(u_H - u) + \cdots .\end{aligned}$$

$$\frac{T}{\Lambda} = -\frac{1}{4\pi} \frac{A'(u_H)}{\Lambda}, \quad \frac{S}{\Lambda^3} = \frac{2\pi}{\kappa^2} \left(\frac{\Sigma(u_H)}{\Lambda} \right)^3$$

POTENTIAL BOUNDARY CONDITIONS

$$S_{\text{theory}}(T) = \pi^4 T^3 + \frac{3\pi^4}{64} T + \dots$$

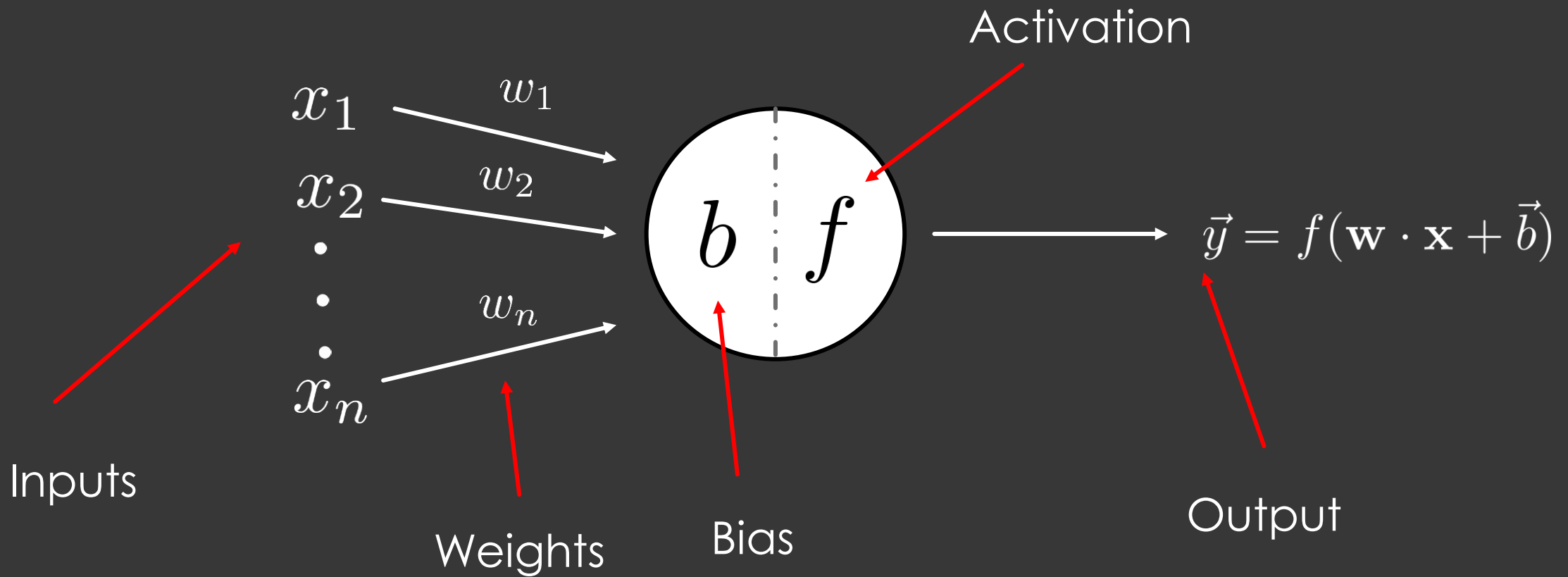
$$C_0 \equiv V(0) + 3 = 0, \quad C_1 \equiv V'(0) = 0, \quad C_2 \equiv V''(0) + 3 = 0.$$

Asymp. AdS

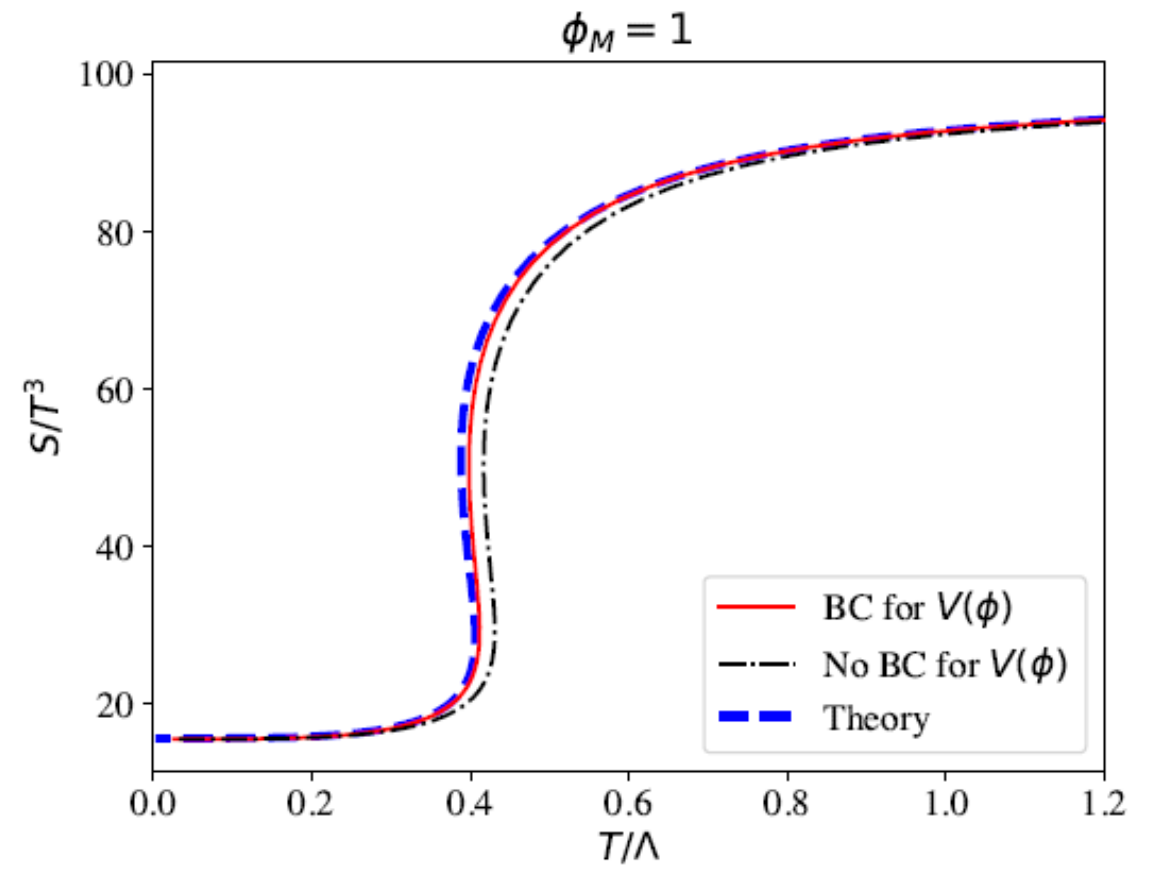
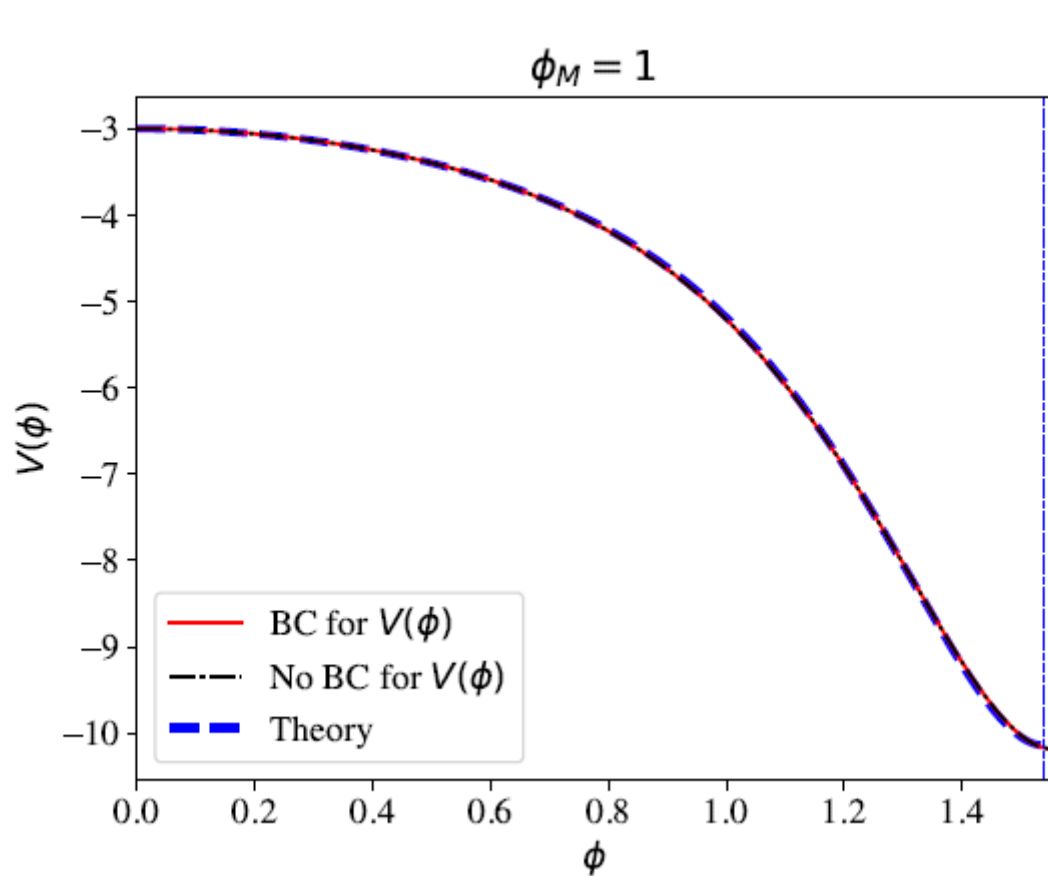
UV fixed point in the
Gauge theory

Fixes the conf. dimension
of the operator

PHYSICS INFORMED NEURAL NETWORKS



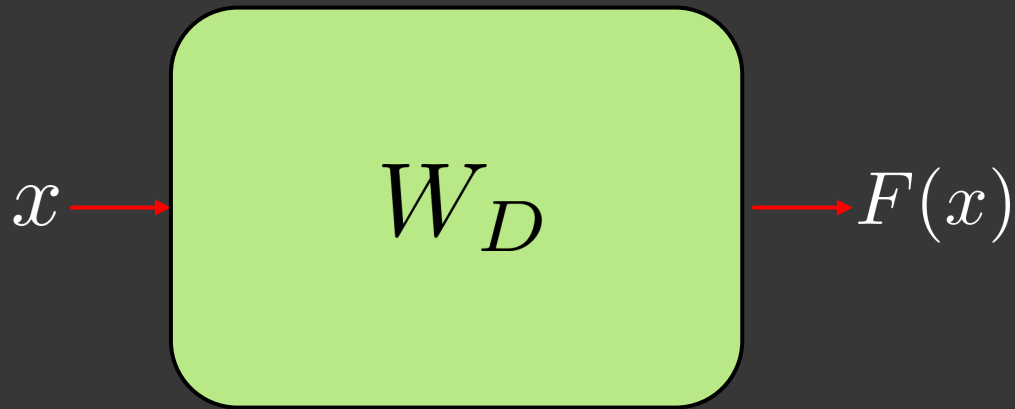
POTENTIAL BOUNDARY CONDITIONS



PHYSICS INFORMED NEURAL NETWORKS

[Dissanayake, M and
Phan-Thien, N. 1994]

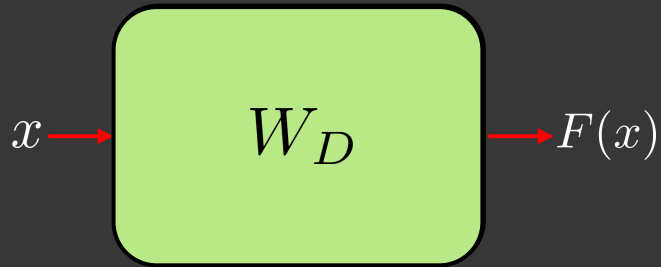
[Lagaris et al. 1997]



[Chen et al. 2020]

- Use a NN for each unknown function
- B.C are fulfilled exactly by construction
- Derivatives are computed analytically
- Slow training but evaluation is instantaneous
- Perfectly suited for inverse problems
- Train for different boundary conditions at once (solution bundle)
- Neurodiffeq

PHYSICS INFORMED NEURAL NETWORKS



Equations

$$E_1 = \nu_\Sigma - \tilde{\Sigma}',$$

$$E_2 = \nu_A - \tilde{A}',$$

$$E_3 = \nu_\phi - \phi',$$

$$E_4 = \nu'_\Sigma + \frac{2}{3}\tilde{\Sigma}\nu_\phi^2,$$

$$E_5 = u^2 \tilde{\Sigma} \nu'_A + \frac{8}{3} V(\phi) \tilde{\Sigma} + \nu_A \left(3u^2 \nu_\Sigma - 5u \tilde{\Sigma} \right) + \tilde{A} \left(8\tilde{\Sigma} - 6u \nu_\Sigma \right),$$

$$E_6 = u^2 \tilde{\Sigma} \tilde{A} \nu'_\phi - \tilde{\Sigma} \frac{dV}{d\phi} + \nu_\phi \left(-3u \tilde{A} \tilde{\Sigma} + u^2 \tilde{\Sigma} \nu_A + 3u^2 \nu_\Sigma \tilde{A} \right),$$

$$E_7 = \left(u \nu_\Sigma - \tilde{\Sigma} \right) \left(u^2 \tilde{\Sigma} \nu_A + 2u^2 \tilde{A} \nu_\Sigma - 4u \tilde{A} \tilde{\Sigma} \right) - \frac{2}{3} u \tilde{\Sigma}^2 \left(u^2 \tilde{A} \nu_\phi^2 - 2V(\phi) \right)$$

Loss function

$$L \equiv \sum_{\alpha} \sum_n \sum_i E_{\alpha} \left(u_n, (T_i, S_i) \right)^2 + V_{BC}.$$

Boundary conditions

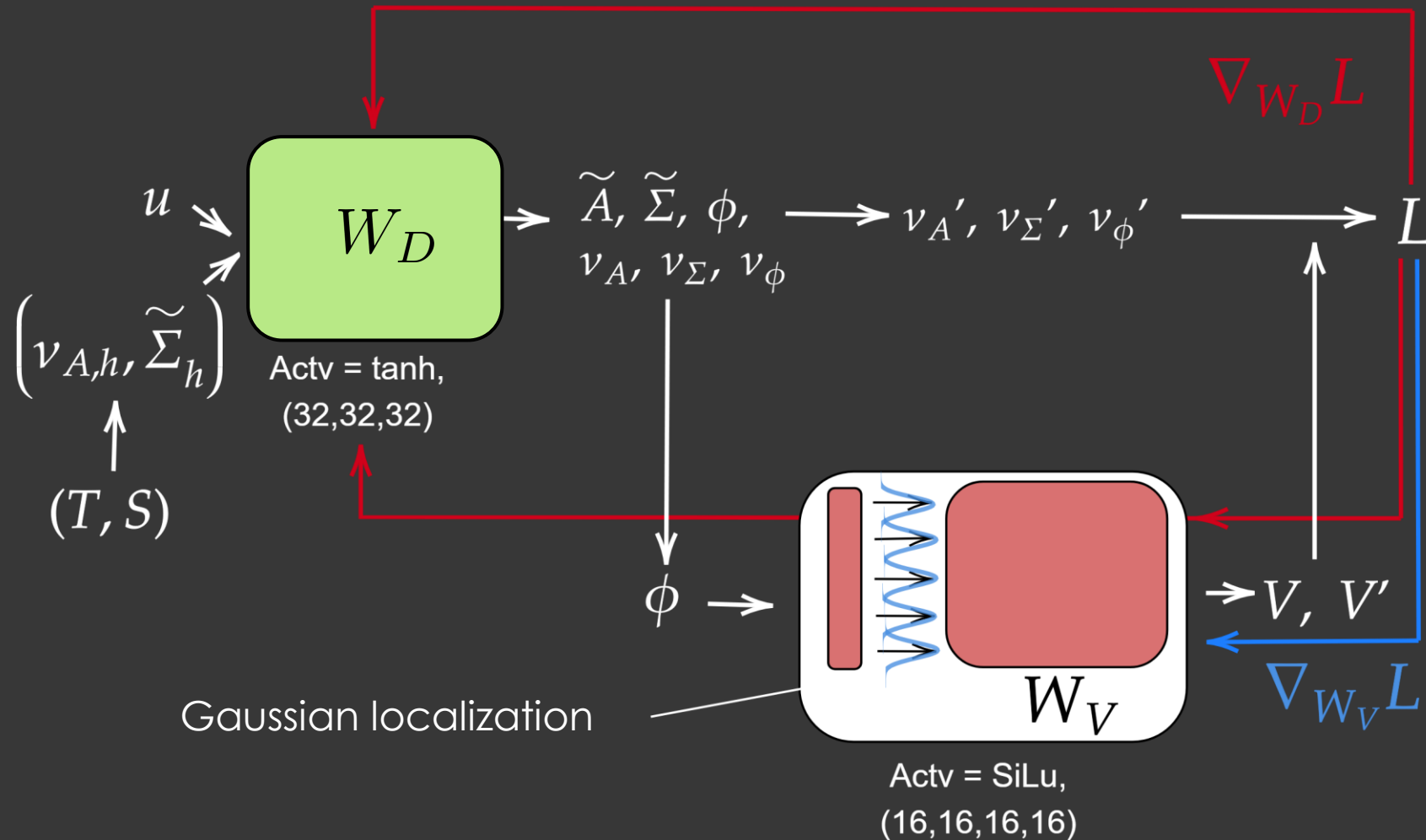
$$\tilde{A}|_{u=0} = 1, \quad \tilde{A}|_{u=1} = 0,$$

$$\tilde{\Sigma}|_{u=0} = 1, \quad \nu_A|_{u=1} = -4\pi T,$$

$$\phi|_{u=0} = 0, \quad \tilde{\Sigma}_{u=1} = (S/\pi)^{1/3}.$$

$$\nu_\phi|_{u=0} = 1,$$

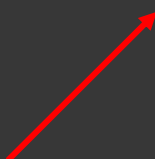
PHYSICS INFORMED NEURAL NETWORKS



FORM OF THE POTENTIAL

Family of potentials with a free parameter ϕ_M

$$V_{\text{theory}}(\phi) = -\frac{4}{3}\mathcal{W}(\phi)^2 + \frac{1}{2}\mathcal{W}'(\phi)^2$$

$$\mathcal{W}(\phi) = -\frac{3}{2} - \frac{\phi^2}{2} - \frac{\phi^4}{4\phi_M^2} + \frac{\phi^6}{10}$$


Regulates the phase transition in the boundary theory

PHYSICS INFORMED NEURAL NETWORKS

Gaussian localization

$$\mathbf{h}_i = \mathbf{h}_{i-1} \cdot \mathbf{W}_{\mathbf{V},i}^T$$

$$\tilde{\mathbf{h}}_1 = \text{SiLU} \left\{ \mathbf{h}_1 \times \exp \left[- \left(\frac{\mathbf{h}_1 - \mu}{\sigma} \right)^2 \right] \right\}$$

PHYSICS INFORMED NEURAL NETWORKS

Gaussian localization

$$\mathbf{h}_i = \mathbf{h}_{i-1} \cdot \mathbf{W}_{\mathbf{V},i}^T$$

$$\tilde{\mathbf{h}}_1 = \text{SiLU} \left\{ \mathbf{h}_1 \times \exp \left[- \left(\frac{\mathbf{h}_1 - \mu}{\sigma} \right)^2 \right] \right\}$$

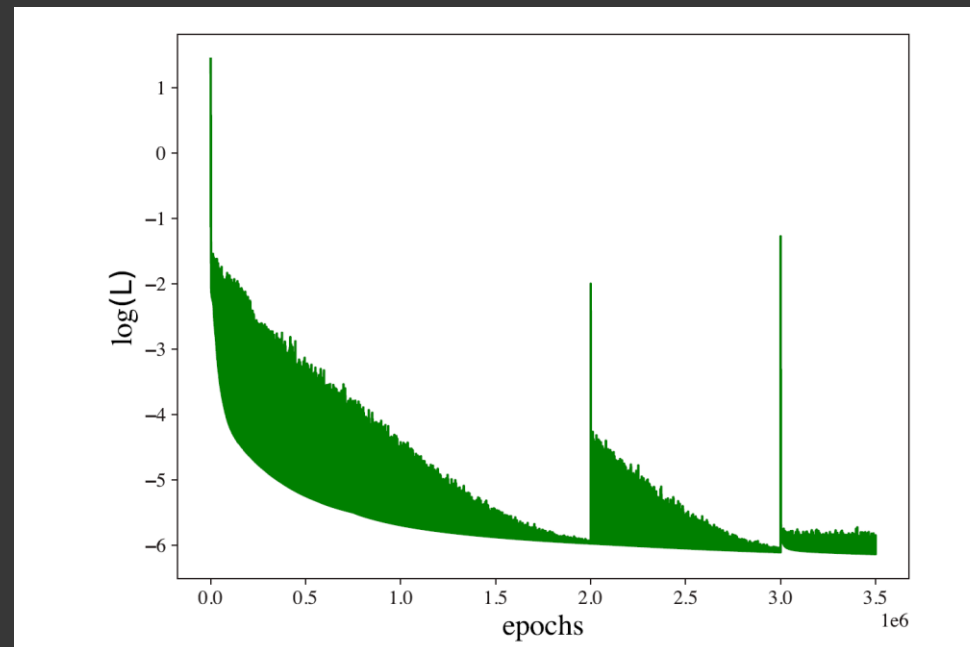
- Fixed sigma's but learnable mu's
- Each neuron of the first layer will specialize on a certain region of the input domain
- Separate loss gradient for inputs that are far apart
- Multi-scale $\rightarrow$ large curvature differences between the boundary and near the horizon region.
- Multi-entangled $\rightarrow$ low-temperature region includes information about the high temperature region

SAMPLING TECHNIQUES FOR THE NN

- The independent variable is sampled using Chebyshev nodes of the second kind. We have used 48 points. This allow us to better constrain the function near the boundaries of the interval.
- The sampling along the $S(T)$ is chosen to have in between 65-70 points.
- This leaves us with a batch size of approx. 3000 points.

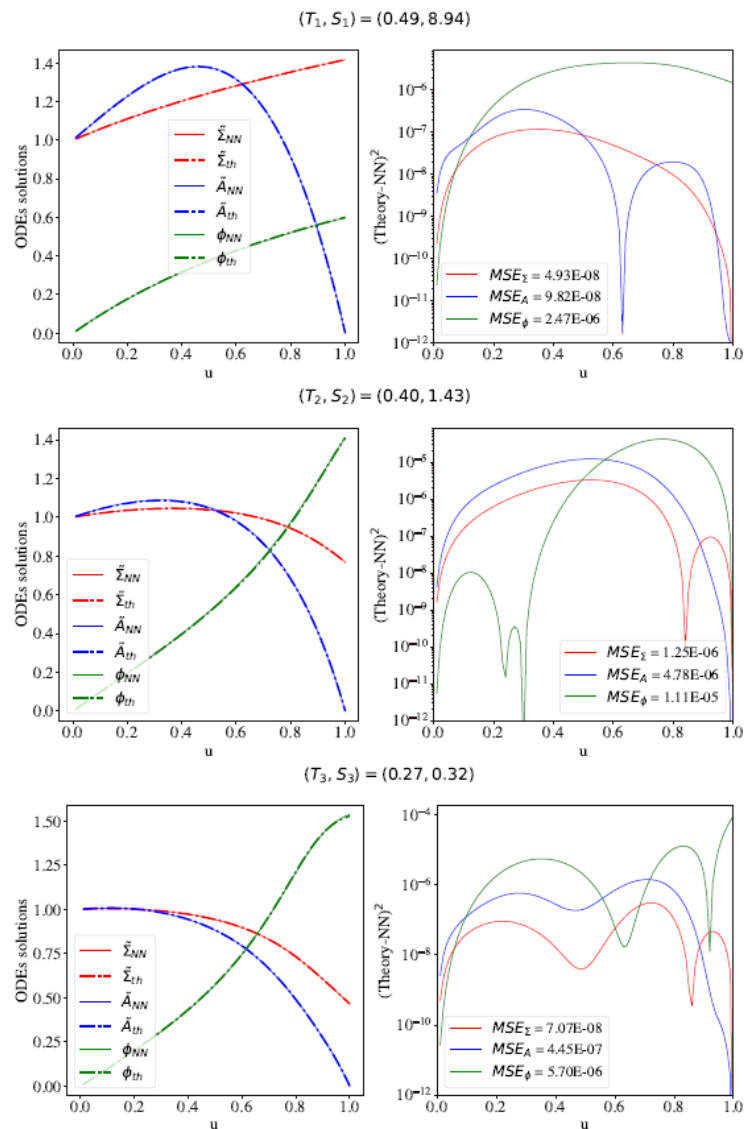
TRAINING DETAILS

- 10 realizations of the same case trained 1e6 epochs.
- The case with the lowest loss is trained until the results are good enough.
- We add the b.c. of the potential as an additional loss and keep training for some epochs until they are fulfilled.

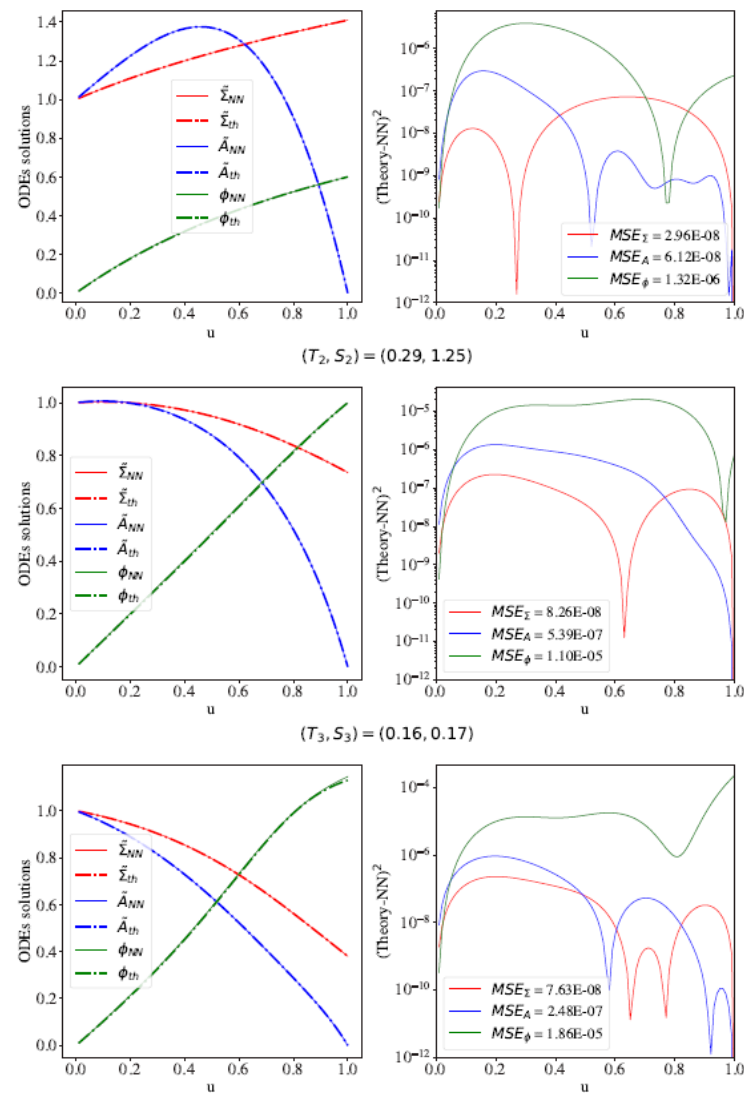


ODE SOLUTIONS

$$\phi_M = 1.0$$

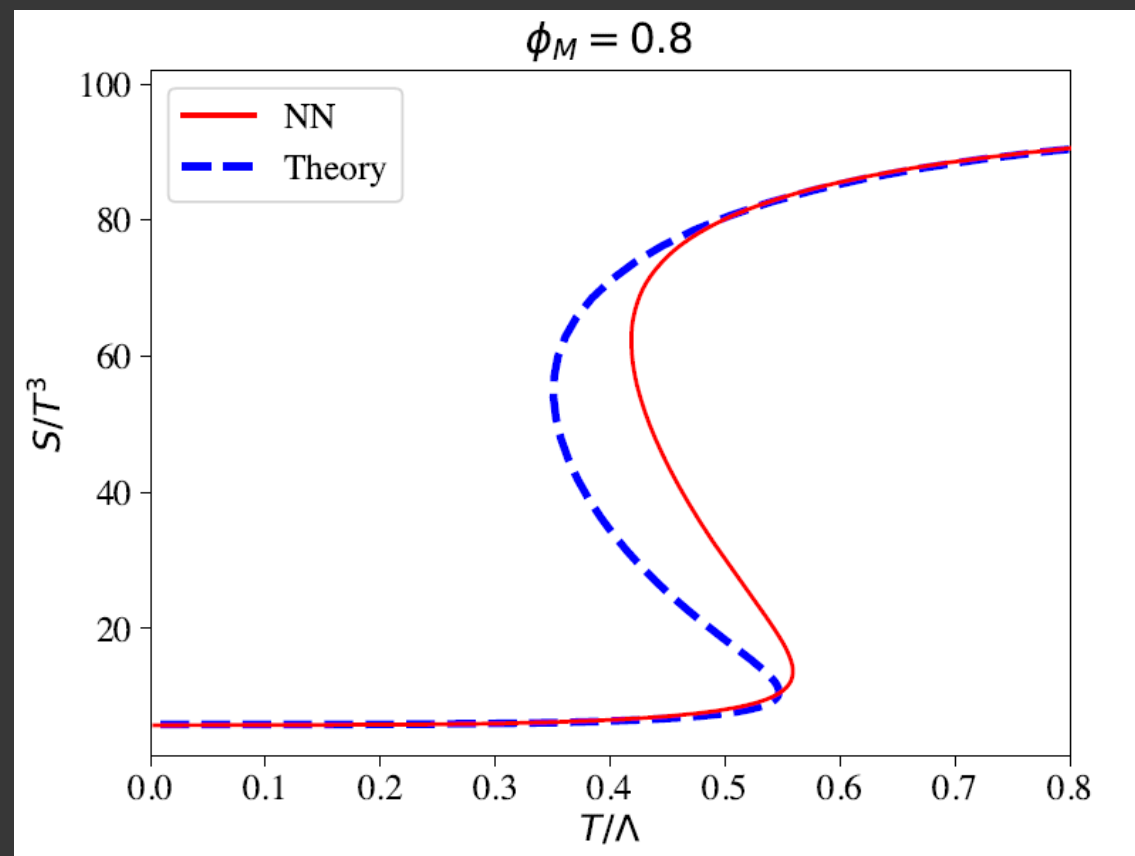
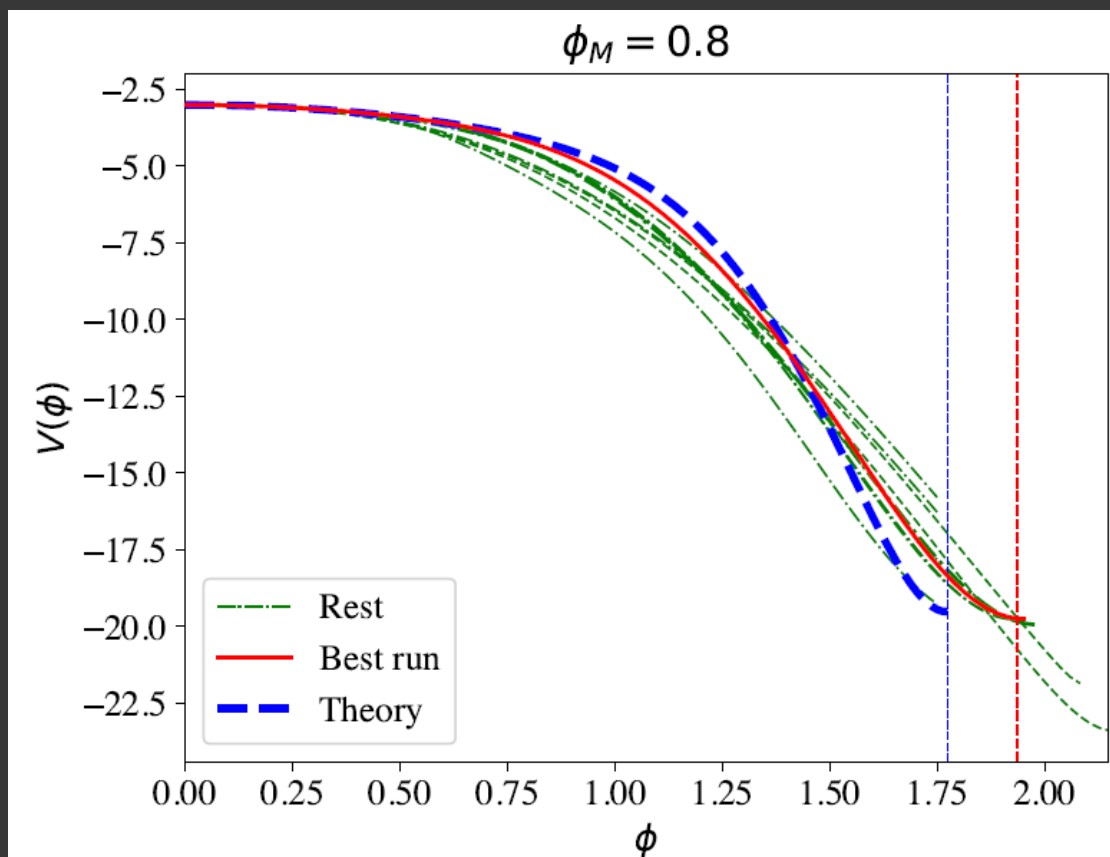


$$(T_1, S_1) = (0.48, 8.80)$$

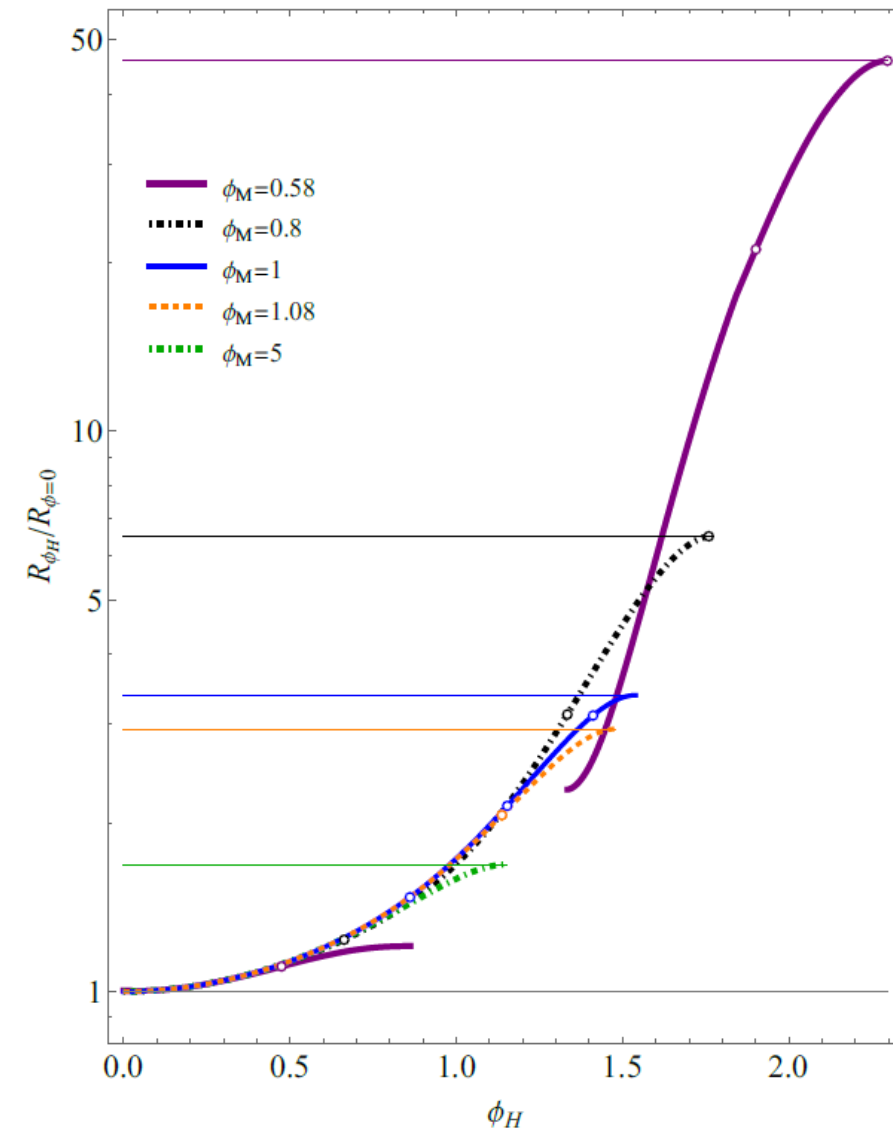


$$\phi_M = 5.0$$

PATHOLOGICAL CASES



MULTI-SCALE OF THE PROBLEM



MULTIHEAD APPROACH

